GEK-83823

DRIVE SYSTEM INSTRUCTIONS

77-mm SILCOMATIC® Plus

Static DC Power Converters

6-Pulse & 12-Pulse

DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC

ISSUE DATE: November 1984

The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met during installation, operation, or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

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SAFETY SYMBOL LEGEND

WARNING

Commands attention to an operating procedure, practice,condition, or statement, which, if not strictly observed, could result in injury or death.

CAUTION

Commands attention to an operating procedure, practice, condition, or statement, which, if not strictly observed, could result in damage to, or destruction of, equipment.

NOTE

Commands attention to an essential operating or maintenance procedure, condition, or statement which must be highlighted.

WARNING

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE, OR MAINTAIN THIS EQUIPMENT.

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

TO MINIMIZE HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE ADHERED TO.

CAUTION

THIS EQUIPMENT IS DESIGNED TO WORK IN A NORMAL INDUSTRIAL ATMOSPHERE. OPERATING RADIO TRANSMITTERS NEARBY MAY CAUSE THE CONTROL EQUIPMENT TO MALFUNCTION. IT IS RECOMMENDED THAT RADIO TRANSMITTERS NOT BE OPERATED WITHIN THE VICINITY OF THE CONTROL EQUIPMENT WHEN THE PANEL DOORS ARE OPEN, NOR AT ANY TIME WITHIN THE CONFINES OF A POWER CONTROL ROOM.

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CHAPTER 1
INTRODUCTION

1-1. PURPOSE AND SCOPE

This instruction book provides information to install, tune up, operate, and maintain 77-mm SILCOMATIC® Plus 6-pulse and 12-pulse power converters. Prior knowledge of electronics and experience in electronic technology is necessary to implement these instructions. Additional information and instructions on process control, interface cabling, etc., are required by the user. This information, including specific power converter elementary diagrams, is supplied with each requisition.

These instructions do not purport to cover every contingency to be met during installation, maintenance, or trouble-shooting. Should further information be desired or particular problems arise which are not sufficiently covered here, they should be referred to General Electric Company.

Recommended changes, additions or deletions to supplied documentation should be addressed to:

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Salem, Virginia 24153

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CHAPTER 2
SYSTEM DESCRIPTION

2-1. INTRODUCTION

This section describes typical configurations of both 6-pulse and 12-pulse 77-mm SILCOMATIC® Plus power converters. Basic operation is described using a block diagram. Detailed circuit descriptions are provided in Chapter 6.

2-2. FUNCTIONAL DESCRIPTION

The 77-mm SILCOMATIC Plus power converter is supplied in both 6-pulse and 12-pulse configurations. Because the 12-pulse system is basically two parallel-connected 6-pulse systems, the following description, while describing specifically the 6-pulse configuration, can be extended to the 12-pulse configuration.

2-2.1 Six-Pulse Configuration

The 77-mm SILCOMATIC® Plus Power Converter is a complete rectifier system which converts ac power to dc power for motor drive applications. The major system elements are shown in a simplified block diagram in Figure 2-1.

1. Main Power Disconnect (Customer-supplied): Provides transformer primary protection and power circuit isolation.

2. Rectifier Transformer: provides magnetic isolation, and coordinates SCR bridge voltage and phase requirements with the available ac power source.

3. Line-to-Frame Filters: Attenuate voltage transients capacitively coupled to transformer secondary winding upon closure of the high voltage circuit breaker. This provides electrical noise control and protection for the SCRs.

4. Surge Suppressor: Absorbs the electrical energy released by current chopping (sudden shut-off of current) of the transformer excitation current when the main power disconnect is opened. This protects the SCRs from excessive voltage transients.

5. Current Transformer (CT): The main CTs and auxiliary CTs provide load current sensing for regulator feedback, metering, and overcurrent detection.

6. Thyristor Power Module: Converts ac input power to controlled dc output power. The gating (firing) of the thyristors in the 6-pulse, full-wave power module is controlled by the firing module. A maximum of four thyristor power modules may be paralleled to provide increased drive output power.

7. Signal Electronics Module (SEM): This module contains nearly all of the electronic circuitry for the power converter with the exception of the power supply and the firing module. This circuitry is packaged on eight printed wiring boards.

8. Firing Module (FM): The firing module converts the SCR gating signals from the SEM into current signals that drive the SCR gates. The FM contains six HPTN cards, each of which gates one forward and one reverse SCR cell. One HPTN card can gate up to 4 parallel-connected cells.

9. 70V Power Supply: Supplies 70V to the firing module for use in SCR gating.

10. Signal Electronics Power Supply: Provides +5Vdc, +15Vdc, -15Vdc and +28Vdc for electronic circuitry.

11. RLO Relay: Provides a control signal to customer-supplied switchgear to trip or lock out the primary power source.

12. Ripple Reactor (Optional): Provides extra filtering of load current to minimize commutation and heating problems in certain motor applications.

13. Attenuator Board (NATA): Attenuates the dc and ac bus voltages for use in the SEM.

14. DC Breaker: Provides high speed interruption of dc load current in the event of a dc shoot-through during inversion.

    There are two optional arrangements for the dc breaker compartment:

    - The standard configuration will be a Power Break dc breaker in each compartment.

    - An ITE Breaker for each converter may be used in larger drives, mounted either in the breaker compartment or separately from the converter.

15. Temperature Sensor Card (NTSC): Develops a voltage that is proportional to the ambient temperature. This information is used in the calculation of the SCR junction temperature by the thermal model.

2-2.2 Twelve-Pulse Configuration

As explained above, the 12-pulse power converter comprises two parallel-connected 6-pulse converters, with additional control circuits to coordinate normal converter operation and maintain current balance. Twelve-pulse ripple (output) is obtained by phase shifting the bridge power of one converter with respect to the other 30 electrical degrees. Paralleling is accomplished through the use of interphase reactors. Refer to Figure 2-2.

CURRENT BALANCE. Both Signal Electronics Modules (SEMs) A and B have their own current balance software routine. This routine looks at two signals: current feedback (IFK) and average current (IAVE). The routine adjusts each converter's output current so that the IFK and IAVE signals are equal.

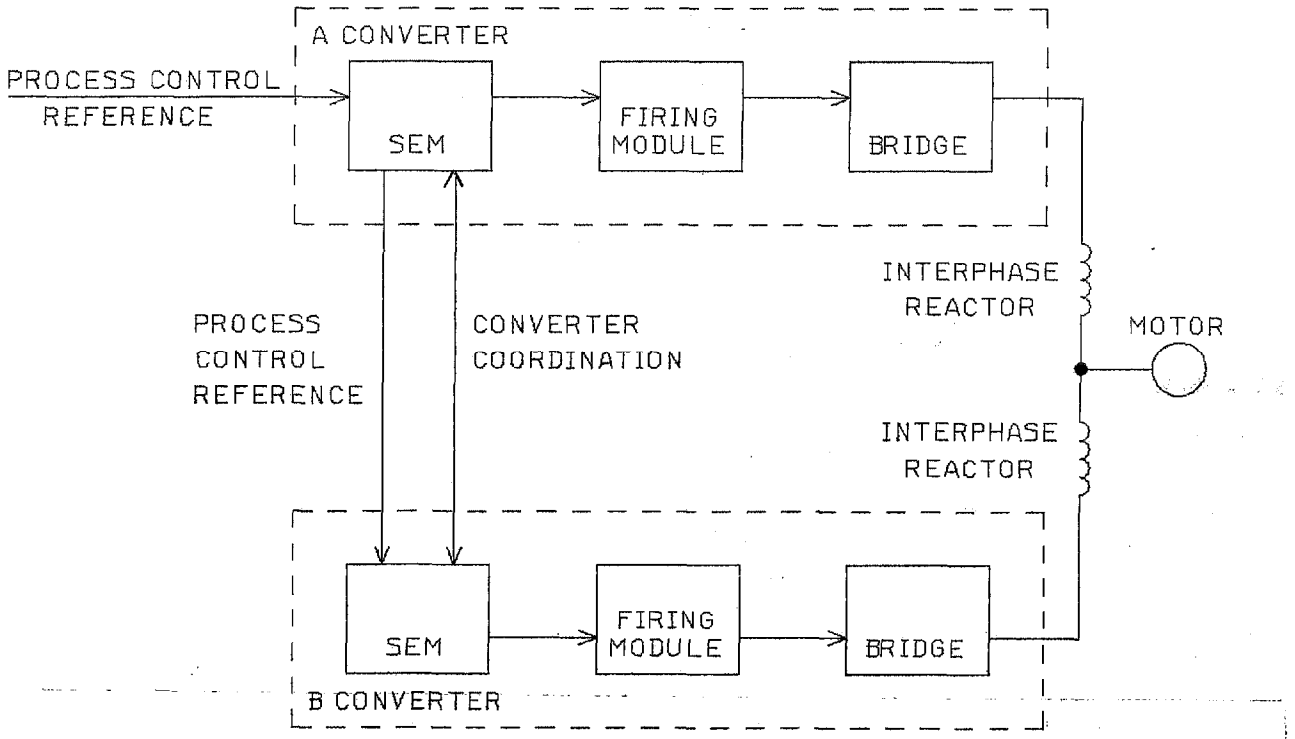


Figure 2-2.
Twelve-Pulse Combined Power Converter Block Diagram

CONVERTER COORDINATION. This is achieved through a 20-pin ribbon cable running between SEMs. The process control reference and suicide signals enter SEM A and are fed to SEM B through the ribbon cable. Coordinated reversal for the two converters is achieved by establishing SEM A as the "Master" and SEM B as the "Slave". SEM A initiates reversal, when required by the process control, by ordering SEM B to reverse. When SEM B reverses it notifies SEM A, which then reverses. Each SEM feeds its Silco fault signal (SFLT), current feedback signal (IFK), and voltage feedback signal (VFK) to the other through the ribbon cable. Each converter also sends its own SFLT signal back to the process control.

2-3. PHYSICAL DESCRIPTION

2-3.1 Six-Pulse Configurations

A typical 6-pulse power converter is shown in Figure 2-3. A maximum of four thyristor power modules may be paralleled to provide increased output power.

There are two optional arrangements for the dc breaker compartment:

- o The standard configuration is a Power Break dc breaker.
- o An ITE Breaker may be used in larger drives, mounted either in the breaker compartment or separately from the converter.

2-3.2 Twelve-Pulse Configurations

As explained previously, a 12-pulse converter is basically two interconnected 6-pulse converters (shown in Figure 2-3). The floor plan of a typical power converter is shown in Figure 2-4. A maximum of four thyristor power modules may be paralleled to provide increased output power.

There are two optional arrangements for the dc breaker compartment:

- The standard configuration will be a Power Break dc breaker in each compartment.
- An ITE Breaker for each converter may be used in larger drives, mounted either in the breaker compartment or separately from the converter.

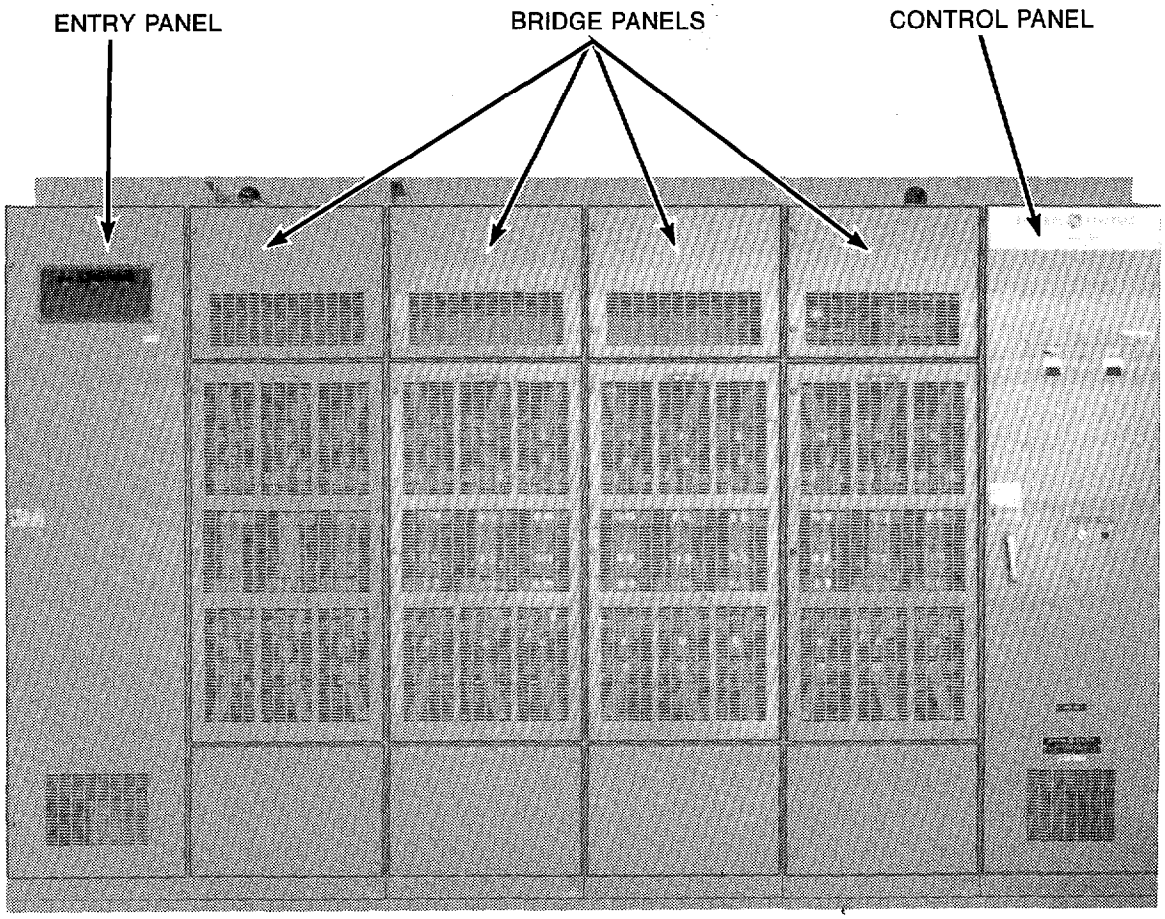


Figure 2-3. A Typical 77-mm SILCOMATIC® Plus 6-Pulse Power Converter

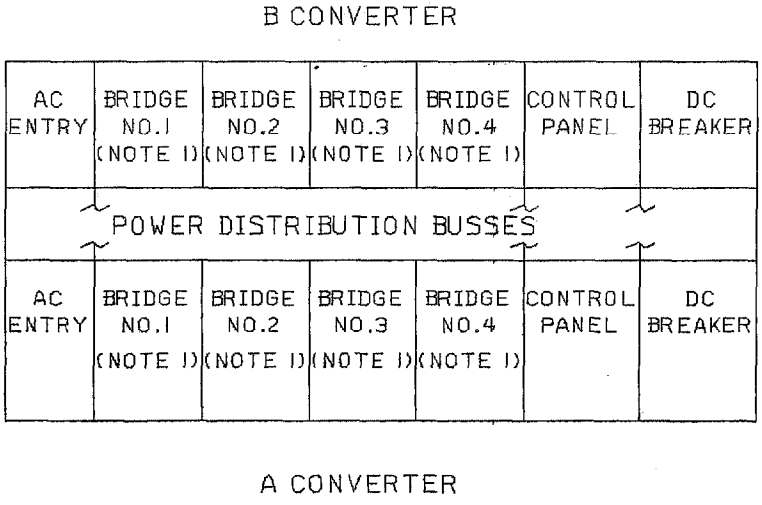


Figure 2-4.
A Typical 77-mm SILCOMATIC® Plus 12-Pulse Power Converter

CHAPTER 3
GENERAL REQUIREMENTS FOR RECEIVING, HANDLING AND STORAGE

3-1. RECEIVING EQUIPMENT

All items of equipment, before leaving the factory, are carefully inspected and packed by personnel experienced in the proper handling and packing of electrical equipment. Upon receipt of any apparatus, an immediate inspection should be made for any damage sustained while enroute. If damage or rough handling is evident, a claim for damage should be filed with the carrier at once. Both the transportation company and General Electric Company should be notified promptly. Information as to damaged parts, part number, case number, requisition number, etc., should accompany the claim.

The equipment should be placed as close as possible to its permanent location, or at its storage location, before unpacking. Unpacking should be done with care to prevent marring or damaging parts.

Some parts, such as bolts, screws, etc., are packed in special containers to keep them together until all parts have been removed from the packing material and quantities checked.

The equipment may be stored in the same environment in which it was designed to operate. Equipment built for operation in an outdoor environment may be stored outdoors and equipment built for operation in an indoor environment must be stored indoors (refer to storage section for additional storage preparation). Indoor equipment should be placed under adequate cover immediately upon receipt because packing coverings are not suitable for temporary out-of-doors or unprotected storage.

Any items removed from the equipment for storage preparation (or any other reason) must be replaced prior to placing the equipment in service.

3-2. HANDLING

Equipment is generally shipped by truck. Small parts or modules may be shipped by air. Vibrations and shock inherent in rail shipment make it the least preferred shipping method.

Refer to equipment outline drawings for preferred methods of lifting and moving the equipment.

The equipment's center-of-gravity should always be carefully observed so that the equipment will not fall.

3-3. STORAGE OF EQUIPMENT

Only authorized personnel should handle and prepare the equipment for storage. All equipment should be secured against entry by unauthorized personnel.

If any part of the equipment is not to be installed immediately, it must be stored in a clean, dry place and protected from sudden changes in temperature and humidity. If the temperature of the storage room varies to such an extent that equipment surfaces are exposed to sweating or freezing conditions, it must be protected by a safe, reliable heating/cooling system to minimize corrosion by gaseous contaminants (activated by moisture). The internal temperature of the equipment should be kept 5°C to 10°C above that of the storage room or outside temperature (ambient temperature) to eliminate condensation inside the equipment.

For maximum electronic component life during equipment operation, the internal temperature should be maintained between +0° and +40°C. The internal temperature of equipment in storage must be maintained between -40° and +70°C.

Equipment may be stored in environments of 5 - 95% relative humidity with the temperature conditions as stated above to prevent condensation.

To minimize the accumulation of particulate contaminants, a breathable type cover (canvas, etc.) should be used to cover the equipment while in storage.

Do not use plastic type covers (polyethylene, etc.) without specific provisions to prevent condensation beneath this type cover.

3-3.1. Outdoor Storage

For outdoor storage, an inspection should be made to insure that all container openings are sealed against wind, rain, snow, sleet, flooding, dust, etc.

Heaters, air conditioners and/or air purging systems may be required to meet the above storage criteria. If so, it will be necessary to install and check this equipment prior to its operation.

CAUTION

TO AVOID A FIRE HAZARD, REMOVE ALL CARTONS AND OTHER MISCELLANEOUS MATERIAL PACKED INSIDE UNITS BEFORE USING HEATERS

If equipment has been exposed to moisture, it should be tested and checked by qualified personnel of General Electric Company before being placed into service.

Equipment such as batteries, air conditioners, purge systems, etc., must be inspected and serviced as required by the manufacturer. The control should be inspected monthly for damage or signs of deterioration and corrective action taken when necessary.

Important parts such as jack screws, gears and chains of circuit breaker mechanisms, linkages and moving machine-finished parts must be prepared for storage as specified in each individual device instruction book.

Low voltage ac drawout breakers (up to 600V) remain in the equipment for storage. Low voltage dc drawout breakers (AG-7 type) may be prepared for storage separately. Medium voltage drawout breakers (5KV to 15KV) should also be prepared for storage separately. Refer to the appropriate circuit breaker instruction book.

3-3.2. Special Requirements

There are some cases where special preparation and packaging are required for storage and operation, e.g. - tropical or polar climates. These situations should be referred to General Electric Company for its evaluation and recommendation.

CHAPTER 4

INITIAL START-UP INSTRUCTIONS

4-1. INTRODUCTION

This section describes the initial inspection, start-up, and testing of the 6- and 12-Pulse 77-mm SILCOMATIC Plus power converters.

The following procedures describe the routine, initial tune-up of both 6-pulse and 12-pulse power converters. Should a problem occur while following these procedures, consult General Electric Company for additional instruction.

4-2. PRECAUTIONS

The following precautions forewarn the service engineer against procedures which could nullify important protective functions designed into the system.

1. Do not, unless instructed, remove gating signals from only a portion of the bridges parallel-connected to share the load. The power converter is designed to withstand all fault conditions, provided the full complement of bridges is gated and all protective functions are operative.

2. Do not perform high-voltage insulation measurements on control modules and circuits in which semiconductor devices are present. They can be easily damaged. When measuring the insulation resistance of converter power circuits, short all buses together to avoid inducing damaging potentials across rectifying devices. High-potential testing of the rectifier circuits is neither advisable nor necessary.

3. Use extreme care when working around the rectifier circuits and modules. They are high capacity, low impedance paths and destructive arcing faults could occur by exercising careless procedures.

4. Do not attempt operation of the converter using open or defective ac current transformers.

5. Do not remove or insert printed circuit cards into the Signal Electronics Module (SEM) unless the SEM power is off.

6. Do not apply main power to the equipment unless the equipment frame is grounded.

7. Do not operate radio transmitters near an operating 77-mm SILCOMATIC Plus power converter. The radio frequency (RF) signals generated by the transmitter could induce noise into the unshielded wires within the converter, thereby creating counterfeit fault conditions.

4-3. ELEMENTARY DIAGRAM

The elementary diagram supplied with each 12-pulse power converter is organized into two parts -- one for the A Converter and a second for the B Converter. Each sheet is identified by a common 3-digit number and two alphabetic characters: The first character identifies the converter (A or B), and the second identifies the sheet within the particular converter diagram.

4-4. REFERENCES

Throughout the following procedures are references to sheets of the elementary diagram. Because most of the following procedures are repeated on each converter, the references have been abbreviated to indicate only the sheet within a converter diagram. The complete sheet number is obtained by prefixing the indicated sheet with the appropriate converter designation -- A or B.

For example, assume a reference to sheet T. If starting Converter A, the applicable sheet is nnnAT. Alternatively, if starting Converter B, the applicable sheet would be nnnBT.

4-5. TOOLS AND TEST EQUIPMENT REQUIRED FOR CONVERTER CHECKOUT

1. Digital Multimeter (DMM)

2. Hand-held terminal, catalog no. 278A3393P1 (supplied with the 77-mm SILCOMATIC system).

NOTE

For convenience of use, an RS-232 Compatible Terminal with a data rate of 300 baud, even parity, a 7-bit character, and 2 stop bits may be substituted for the hand-held terminal.

3. Test load capable of conducting a minimum current of 50 amperes at rated voltage.

4. Strip chart recorder, with a minimum of two channels. (an 8-channel recorder with digital memory, such as the Gould 2000W Series, is recommended.)

4-6. LOCATION OF MODULES, WIRING BOARDS, AND COMPONENTS

Figure 4-1 shows the locations of modules and certain components in a typical 77-mm SILCOMATIC Plus Power Converter. Only those parts referenced in the following start-up procedure are identified. Additional illustrations in the start-up procedure detail the Control Panel (Figure 4-2), the Signal Electronics Module (Figure 4-3), and the Firing Module (Figure 4-4).

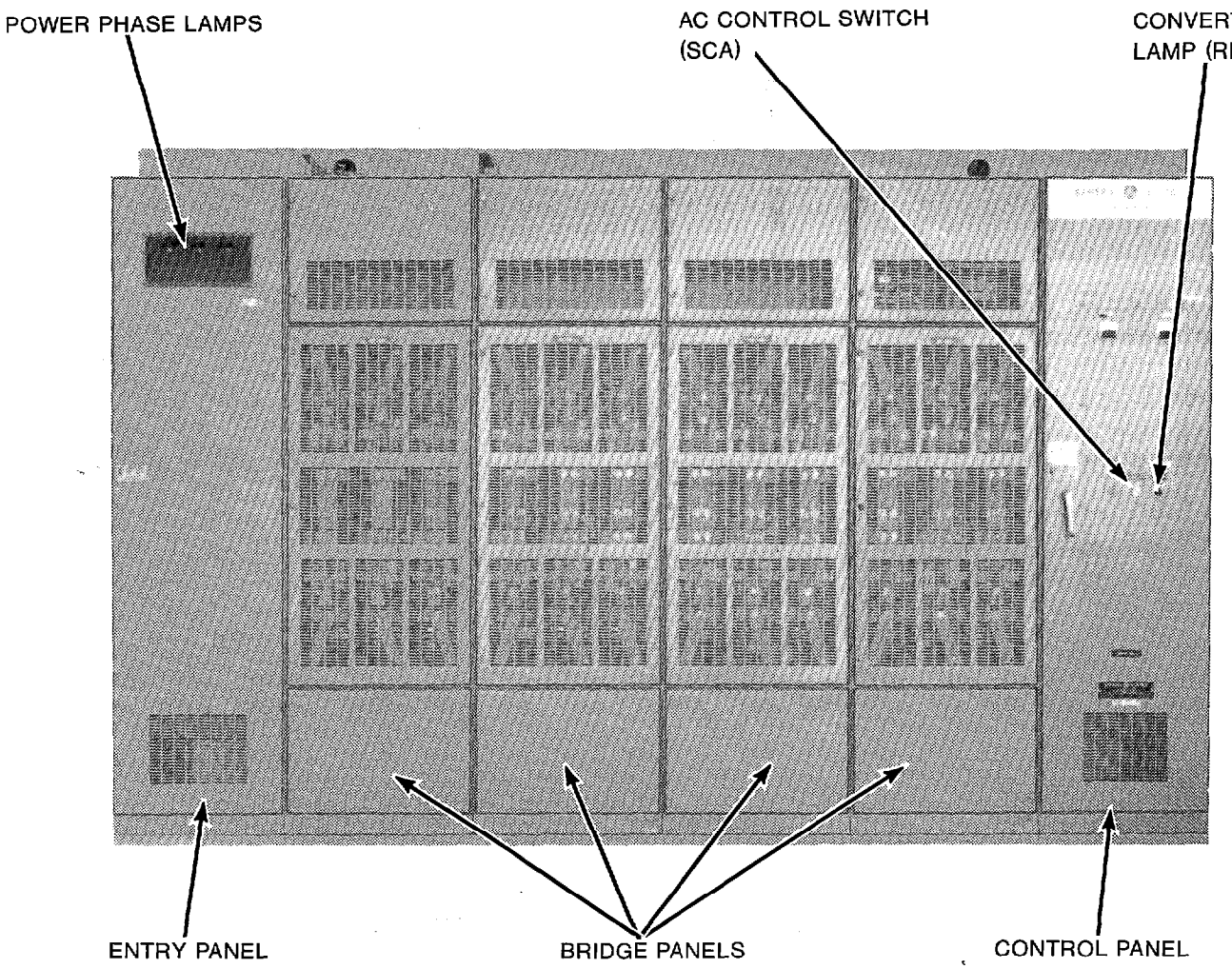


Figure 4-1(a). Location of Modules & Components — Exterior View

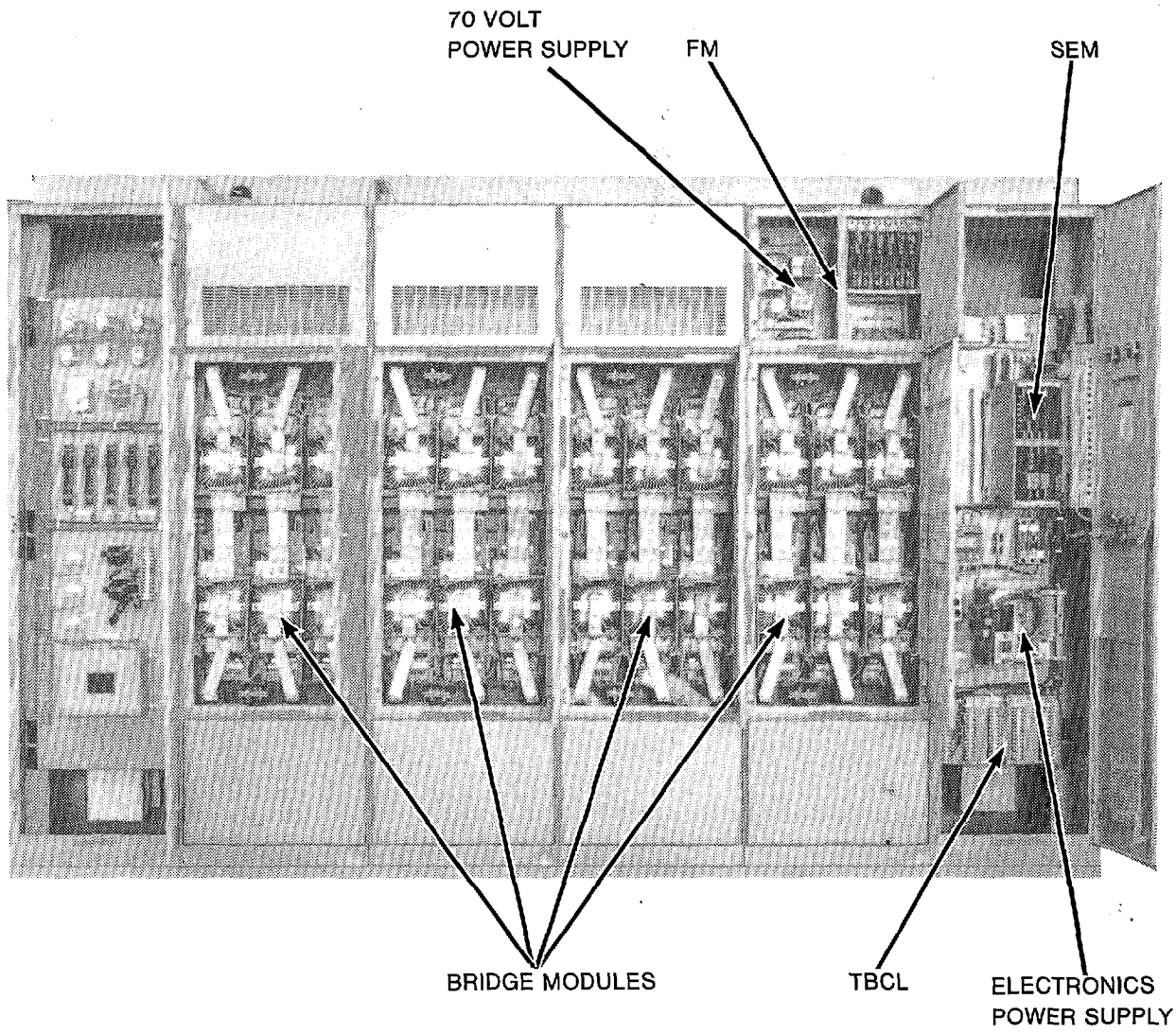


Figure 4-1(b). Location of Modules & Components — Interior View

4-7. INSPECTION OF NEW POWER CONVERTERS

1. Inspect for proper assembly, including mechanical and electrical connections.
2. Check for loose, missing or damaged parts.
3. Perform primary breaker (ac) and converter transformer inspection and tests. Set transformer taps as required to provide rated voltage to the rectifier.
4. Perform insulation resistance tests only on the transformer, between the transformer and power converter, and between the power converter and motor.
5. Check interface wire and cables for proper size and insulation levels (especially grounding leads).
6. Clear and check fans for easy rotation (without power).

4-8. INTERFACE WIRING CHECKS

Check the wiring of the following circuits and repair any open or short circuits.

1. The Overtemperature (and Low Oil, if used) switch circuit to the transformer.
2. The remote Suicide circuit.
3. The remote Reference circuit.
4. The remote Enable circuit.
5 The DC Breaker Closed signal from the dc breaker.
6. The High Speed Trip Ready circuit from the dc breaker (not used on all breakers).
7. The SILCOMATIC Fault (SFLT) circuit to process control.
8. The Power Converter Annunciate (PCA) circuit to process control (if used).
9. The Current Feedback circuit to process control (if used).
10. The Thermal Overload Forward (TOLF) circuit to process control (if used).
11. The Thermal Overload Reverse (TOLR) circuit to process control (if used).

4-9. TEST CONNECTIONS

1. Connect the test load across the output dc bus.
2. Connect the hand-held terminal to SEM connector JB (Figure 4-2).

NOTE

If an RS-232 compatible terminal is used rather than the supplied hand-held terminal, connect it to connector JB and proceed to 4-10

3. Remove Berg jumper BJ4 (on the HFXB card) from the NC posts and reinstall it across the PWR posts. (Refer to Figure 4-3 for the location of the HFXB card.)

4. Set the personality switches, behind the hand-held terminal rear cover, to the following positions:

   Switches 1, 2, & 3 -- up position
   Switches 4, 5, & 6 -- down position

4-10. AUXILIARY CONTROL POWER

1. Ensure phases 1, 2, and 3 on the incoming side of the Control Power Switch (CPS, Figure 4-2) are connected in the proper sequence.

2. Ensure rear panels are in place and doors are shut on SCR panels.

3. Verify that the PSFF power supply circuit breaker (Figure 4-2) is open.

4. Close breaker CPS. Do not turn on bridge power at this time. Check each bridge fan for proper operation and rotation.

5. Using the DMM, measure the voltage between transformer T2 (Figure 4-2) terminals X1 and X2. The voltage must be 115Vac $\pm$ 10Vac.

6. Close the PSFF power supply circuit breaker.

4-11. COMPUTER SELF-TEST

During this test, the internal computer checks the PROM and RAM for compatible revision levels and defective or incorrect parts.

1. Place the HFXA card switches SW1 & SW2 (Figure 4-3) in their center positions and the Thumb Wheel Switch (TWS) at 0.

2. Press and hold HFXA SW2 to start the computer self-test. Make sure that all Light Emitting Diodes (LEDs) on the HFXA and HFPB cards light while SW2 is held. Release SW2 to proceed with the self-test.

3. The computer self-test sequence is detailed in Appendix A. If any of the 18 tests fails, read the message sent to the terminal to identify the defective part. Alternatively, the HFXA and HFPB LEDs can be interpreted, using Appendix A, to determine the defective part.

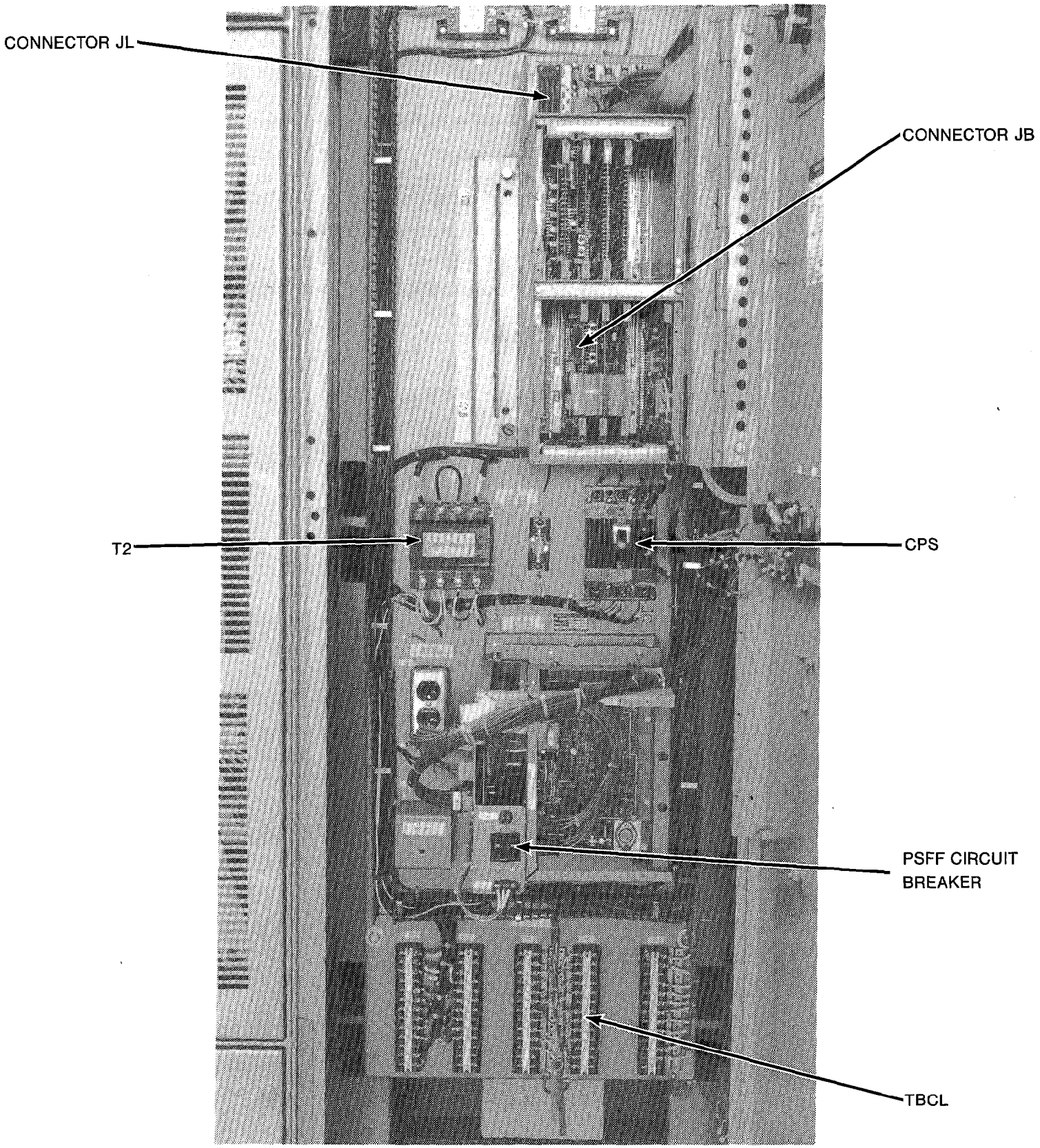


Figure 4-2. Start-Up Test Connections

4. When all tests are completed, a message will be sent to the terminal indicating the computer is OK.

5. When the computer finishes its self-test routine, it automatically starts the bridge test routine. Since there is no bridge power, the computer will fail bridge test and send a fault message to the terminal; ignore this message.

6. Check the IMOK lights on all cards. If any are not on, replace the card.

4-12. CONVERTER-TO-CONVERTER CHECKS (12-Pulse Converters only)

1. Repeat 4-9 through 4-11 on the B-Converter.

2. Reset the converters with switch SW2 (NBIB).

3. Place switches SW1 & SW2 (HFXA) in their upper positions. Check that the PLL LED (NBIB) comes on and that the door-mounted RIL lamps on both converters are off.

4. Apply the ENABLE and SUICIDE* signals.

5. Open the dc breaker on the A-converter.

6. Check that the dc breaker on the B-converter tripped (if option 32 is selected -- which is typical) and that the RIL lamps on both converters are on. Reomve the ENABLE and SUICIDE* signals.

7. Close both dc breakers and reset each converter with switch SW2 (NBIB).

8. Apply the ENABLE and SUICIDE* signals.

9. Open the dc breaker on the B-converter.

10. Check that the dc breaker on the A-converter tripped (if option 32 is selected -- which is typical) and that the RIL lamps on both converters are on.

11. Remove the ENABLE and SUICIDE* signals.

12. Close both dc breakers and reset each converter with switch SW2 (NBIB).

4-13. BRIDGE POWER CHECK

1. Return the drive to the normal operating mode by placing switches SW1 & SW2 (HFXA) in their center positions. Rotate the TWS to position 0. The door-mounted RIL light should be on.

2. Open the dc breaker.

3. Apply bridge power. Check that the PLL LED on the NBIB card has come on.

4. Depress switch SW1 to begin the Bridge Test. When completed, the terminal will display the message BRIDGE TEST COMPLETE: BRIDGE OK.

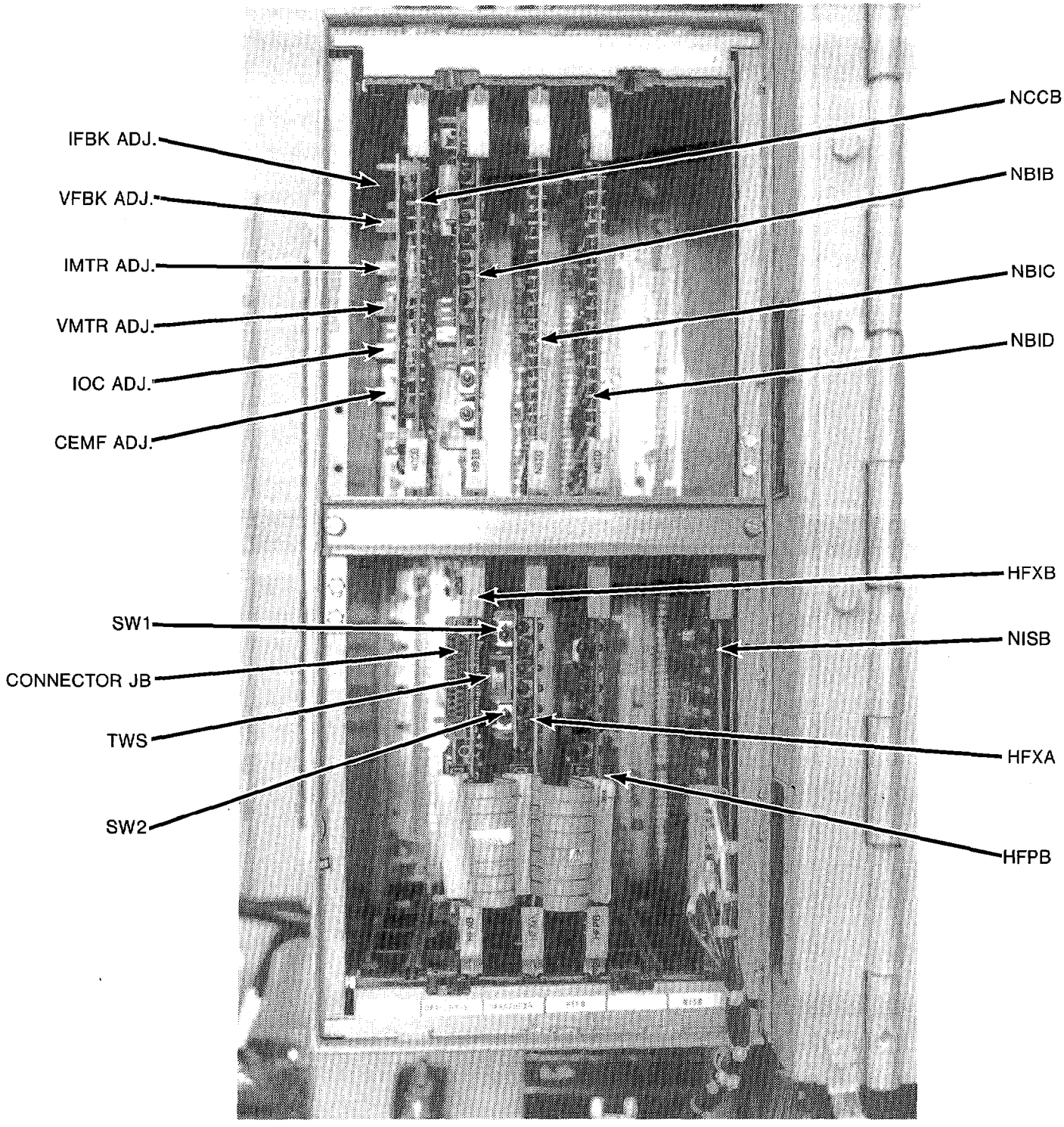


Figure 4-3. SEM Controls & Indicators

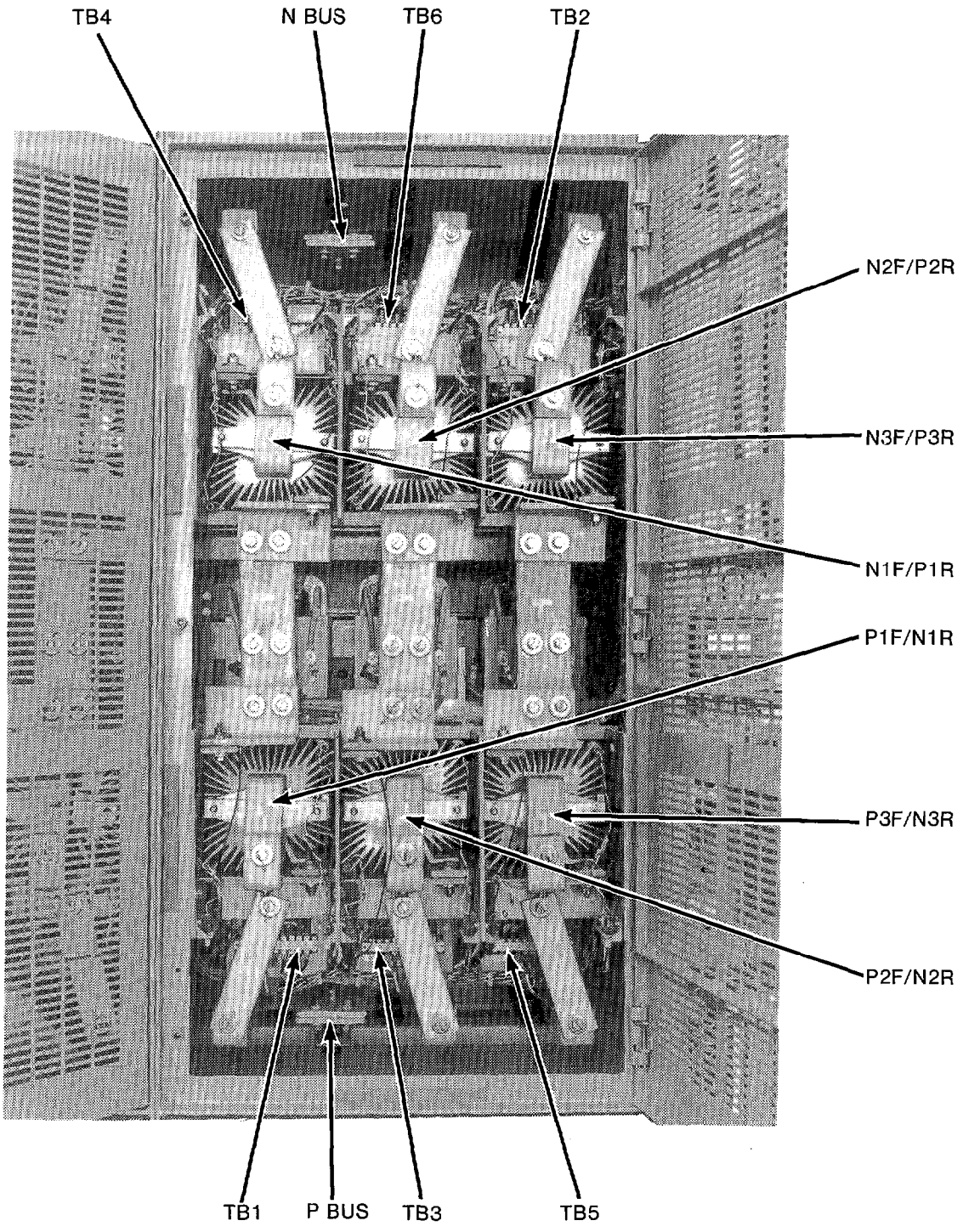


Figure 4-4. Bridge Module

4-14. RESISTIVE LOAD TESTS

Except as noted, the following procedure is to be used for both 6- and 12-pulse converters.

NOTE

The following test steps require a test load across the dc bus. The test load and cabling should be sufficient to carry approximately 50 amperes.

The remainder of this procedure is accomplished by operating the power converter and the motor from the process control regulator.

1. Turn on bridge power to the converter.

2. Close the dc breaker(s). Reset the hardware with switch(es) SW2 (NBIB).

3. Enter a "B" on the terminal. This command makes the computer repeat bridge test.

4. If bridge test is not successful, the terminal will display a message describing the problem.

5. When the bridge test is completed, the terminal will read BRIDGE TEST COMPLETE: BRIDGE O.K.

6. Reset the hardware with switch SW2 (NBIB). The door-mounted RIL lamp should be off. Rotate the TWS to position 0.

7. Check that all LEDs on the HFPB & HFXA cards are off. If they are not, refer to Chapter 8 or the 77MM SEQUENCING LOGIC ELEMENTARY to determine the problem.

8. (12-Pulse Converters only) Disconect the hand-held terminal from the SEM in the A-converter and connect it to the B-converter. Repeat steps 1 through 7.

NOTE

FOR THE REST OF THE RESISTIVE LOAD TEST, AND FOR PART OFTHE LOCKED ROTOR TEST, THE A-CONVERTER AND THE B-CONVERTER BOTH HAVE TO BE SET UP AND OPERATED TEMPORARILY AS SEPARATE 6-PULSE CONVERTERS.

9. (12-Pulse Converters only) Perform the following steps on both converters in order to operate them as individual 6-pulse converters.

   (a) Enter the GE/DSD Sideline Monitor by typing an "M" into the terminal. The password needed is "WDB". The computer should acknowledge this by sending the message GE/DSD SIDELINE MONTOR.

(b) Call up location 3200 and store a "10H" in it. This will set the converter to perform as a 6-pulse drive; however, this action does not destroy the requisition-variable software that is stored in EEPROM.

(c) Now enter a "CTRL C" to return to the 77-mm SILCOMATIC Operating System.

(d) To return each converter to the 12-pulse mode at the conclusion of this procedure, do either of two things:

(1) Reset the software by depressing switch SW2 on the HFXA card in each SEM.

(2) Turn the power supply off and then back on.

NOTE

DO NOT TURN THE POWER SUPPLY OFF UNTIL YOU ARE READY TO RETURN TO THE 12-PULSE MODE OR YOU WILL HAVE TO REPEAT STEPS (a) - (c) TO RESET THE CONVERTERS AS INDIVIDUAL 6-PULSE DRIVES.

10. (12-Pulse Converters only) Open the dc breaker on the B-Converter. This disables the B-Converter so that the A-Converter can be tested independently.

11. Check that the voltage reference is 0 volts.

12. Connect the DMM across the converter P-N bus.

13. Apply the ENABLE and SUICIDE* signals. DO NOT UNSUICIDE THE CURRENT REGULATOR AT THIS TIME, OPERATE THE DRIVE OPEN LOOP ONLY. Check that the dc voltage is 0 volts on the bridge.

14. Verify that the P-N bus voltage can be adjusted through its rated range by varying the process control voltage reference from approximately +8 V dc (corresponding to rated forward voltage) through 0 V to -8 V dc (corresponding to rated reverse voltage).

15. Check that the polarity of the current feedback signal at the Process Control current regulator is correct.

16. Reduce the voltage reference to 0 volts. Remove the ENABLE and SUICIDE* signals.

17. If the SEM is a Form B version, which interfaces with the Digital Regulator when it is used in the Process Control equipment, and the 1GNFLT signal from NISB (slot 1A) is taken to Process Control, perform the following:

(a) Connect a 10 K ohm, 10 W resistor between the P-BUS and ground.

(b) Unsuicide the drive and increase the voltage reference until positive rated dc voltage is reached.

(c) Check that a ground fault was detected at Process Control and that the PGND LED in the NVIA card (SEM slot 1B) came on.

(d) If a ground fault was not detected, adjust potentiometer R3 on the NVIA card until the PGND LED comes on.

(e) Reduce the voltage reference to 0 volts and suicide the drive. Reset the NVIA card by pressing its GND RST switch.

(f) Disconnect the 10 K ohm, 10 W resistor from the P-BUS and connect it from the N-BUS to ground.

(g) Unsuicide the drive and increase the voltage reference until negative rated dc voltage is reached.

(h) Check that a ground fault was detected at Process Control and that the NGND LED in the NVIA card came on.

(i) Reduce the voltage reference to 0 volts and suicide the drive. Reset the NVIA card by pressing its GND RST switch.

(j) Remove the 10 K ohm resistor from the N-BUS and ground.

18. (12-Pulse Converters only) Close the B-converter dc breaker.

19. (12-Pulse Converters only) Open the A-Converter dc breaker.

20. (12-Pulse Converters only) Repeat steps 11 - 16 to test the B-Converter.

21. (12-Pulse Converters only) When finished testing the B-Converter, close the A-converter dc breaker.

4-15. LOCKED ROTOR OR SHORTED RIPPLE REACTOR TESTS

CAUTION

SEVERE DAMAGE TO THE MOTOR OR REACTOR MAY OCCUR IF LARGE VOLTAGES AND CURRENTS ARE PRODUCED FOR MORE THAN A FEW SECONDS AT A TIME.

4-15.1. Preparation for Test

A. Locked Rotor Test:

a. Fabricate and install an "A-frame" in such a location as to immobilize the rotor.

b. Short-circuit the shunt field leads. Ensure either the input power is removed from the field exciter or the field exciter is disconnected from the shorted field.

B. Shorted Ripple Reactor Test:

NOTE

A shorted ripple reactor test is an alternative test which can be used if a ripple reactor is installed on the dc bus.

a. Connect a shorting bar or bus bar across the power converter terminals below the ripple reactor in order to short the motor leads together.

CAUTION

TO AVOID EQUIPMENT DAMAGE DURING THE REMAINDER OF THIS PROCEDURE, THE CURRENT REGULATOR (IN THE PROCESS CONTROL EQUIPMENT) MUST BE OPERATED CLOSED-LOOP.

4-15.2 Test Procedure

1. (12-Pulse Converters only) Open the dc breaker for the B-Converter. This disables the B-converter so that the A-converter can be tested independently.

2. Connect a digital multimeter (DMM) to measure the voltage across the dc shunt in the motor armature circuit.

3. Determine suitable forward and reverse motor currents at which to check the current feedback calibrations, preferably rated motor current.

4. Determine the voltage that will be across the dc shunt whenever the selected forward or reverse current is applied.

5. Apply the ENABLE and SUICIDE* signals. Increase the reference until the dc shunt voltage indicates that the desired forward motor current has been reached.

6. Check that the voltage magnitude and polarity on TP7 (NCCB) are correct for the motor current selected.

7. Check that the current feedback voltage and polarity at the Process Control regulator are correct for the motor current selected.

8. Reduce the voltage reference to 0 volts.

9. Increase the voltage reference until the dc shunt voltage indicates that the desired reverse motor current has been reached.

10. Check that the voltage magnitude and polarity on TP7 (NCCB) are correct for the motor current selected.

11. Check that the current feedback voltage and polarity at the Process Control regulator are correct for the motor current selected.

12. Reduce the voltage reference to 0 volts. Remove the ENABLE and SUICIDE* signals.

NOTE

If you are operating the A-Converter, the process control regulator should be tuned at this time. The ripple method that is used to tune the regulator is only valid for a 6-pulse ripple; therefore, a 12-pulse drive has to be tuned up with only one of its converters operating in the 6-pulse mode.

13. Refer to the elementary sheet S for the Instantaneous OverCurrent setting (IOC) of the drive (this is also known as the Suppression level).

14. Determine the dc shunt voltage that will be developed for the IOC current.

15. Apply the ENABLE and SUICIDE* signals. Increase the voltage reference until the dc shunt voltage indicates that the forward IOC current level has been reached.

16. Check that the dc breaker on the Silcomatic tripped. Reduce the voltage reference to 0 volts. Remove the ENABLE and SUICIDE* signals.

17. Check, on the faulted converter, that the terminal message reads INSTANTANEOUS OVERCURRENT DETECTED.

18. If the drive does not trip at the IOC level, adjust potentiometer IOC (NCCB) until it does.

19. Reset the drive with switch SW2 (NBIB). Check that the door-mounted RIL lamp is off. Close the dc breaker.

20. Repeat steps 14 - 19 for the reverse IOC check. If the converter under test is a 6-pulse unit, proceed to step 36.

21. (12-Pulse Converters only) Close the B-Converter dc breaker.

22. (12-Pulse Converters only) Open the A-Converter dc breaker.

23. (12-Pulse Converters only) Repeat steps 2 - 20 to test the B-Converter.

24. (12-Pulse Converters only) When finished testing the B-Converter, close the A-converter dc breaker.

25. (12-Pulse Converters only) Return both converters to the 12-pulse mode by doing either of the following:

(1) Reset the software by depressing switch SW2 on the HFXA card in each SEM.

(2) Turn the power supply off and then back on.

26. (12-Pulse Converters only) Verify that the drive is back in the 12-pulse mode, both dc breakers closed, and that both converters are reset so that both door-mounted RIL lamps are off.

27. (12-Pulse Converters only) Connect a strip chart recorder to TP7 (IFK)(NCCB) in the A-Converter and TP7 (IFK)(NCCB) in the B-Converter. These two signals are the individual current feedback signals.

28. (12-Pulse Converters only) Apply the ENABLE and SUICIDE* signals. Increase the reference until rated 12-pulse current is attained.

29. (12-Pulse Converters only) Check that signals IFK(A) and IFK(B) are within 5% of each other. If they are not balanced, record which is larger.

30. (12-Pulse Converters only) Reduce the reference to 0 volts and suicide the drive. If signals IFK(A) and IFK(B) are balanced, skip to step 36.

31. (12-Pulse Converters only) Refer to elementary sheets AT and BT.

32. (12-Pulse Converters only) The 12-pulse Current Balance Regulator Gain (memory location 3C0E), shown on the elementary for each converter, will have to be modified to balance the two currents.

33. (12-Pulse Converters only) Decrease the gain of the converter with the greater current and increase the gain of the converter with the lesser current.

CAUTION

The gain of the regulators should never exceed 10H to achieve converter balance.

34. (12-Pulse Converters only) Repeat steps 28 - 33 until the two converter currents balance.

35. (12-Pulse Converters only) If the Current Balance Regulator Gains were changed, record on the elementary the values that were chosen and store them into nonvolatile RAM using the following procedure:

A. Place a jumper between HFXB TP4 & TP5.
B. Call up location 3200 and store an "8" in it. The computer should confirm the entry with the message NONVOLATILE RAM UPDATED. This action stores the changed requisition data into nonvolatile memory.

C. Contact the design engineer through Product Service, DSD Salem, VA, and inform him of the changes.

NOTE

Omit step 36 if the SILCOMATIC Power Converter uses only a single bridge panel

36. Bridge Current Sharing Check:

**** REPEAT THE FOLLOWING STEPS A - E FOR ONE BRIDGE OF EACH DRIVE. ****

A. Disconnect one forward gate lead on the PH1 bus.

B. Apply the ENABLE and SUICIDE* signals. Increase the voltage reference in the forward direction until the drive faults on Current Imbalance (about 100-300 amps will fault the drive). Current should go to 0. Reduce the voltage reference to 0 volts.

C. Remove the ENABLE and SUICIDE* signals.

D. A message should be displayed on the terminal, identifying the non-conducting leg.

E. Reconnect the disconnected gate lead.

F. Repeat A thru E for phases 2 and 3.

4-16. MOTOR TEST

1. Verify the motor field is properly connected to the exciter (not shorted) and that input power is applied to the exciter.

2. Verify that any shorts across the motor leads have been removed and the rotor is not locked.

3. Apply the ENABLE and SUICIDE* signals; DO NOT UNSUICIDE THE CURRENT REGULATOR AT THIS TIME, OPERATE THE DRIVE OPEN LOOP ONLY. Check that the dc voltage is 0 volts on the bridge.

4. Slowly increase the voltage reference to accelerate the motor to one-half forward base speed and one-half rated voltage.

5. Check that the door-mounted voltmeter is reading correctly. If not, adjust potentiometer VMTR (NCCB) until it does.

6. (12-Pulse Converters) Repeat step 5 on the B-converter.

7. Check that the voltage magnitude and polarity at TP8 (NCCB) are correct for the dc bus voltage present. If not, adjust potentiometer VFBK (NCCB) until it is.

8. (12-Pulse Converters) Repeat step 7 on the B-converter.

9. Check that the voltage magnitude and polarity at TP15 (NCCB) are correct for the dc bus voltage present. If not, adjust potentiometer CEMF (NCCB) until it is.

10. (12-Pulse Converters) Repeat step 9 on the B-converter.

11. Check that the tachometer feedback into the Process Control speed regulator is correct in both magnitude and polarity for the speed of the motor.

12. Slowly change the voltage reference until one-half reverse base speed and one-half rated voltage on the motor are attained.

13. Check that the tachometer feedback into the Process Control speed regulator is correct in both magnitude and polarity for the speed of the motor.

14. Reduce the voltage reference to 0 volts. Remove the ENABLE and SUICIDE* signals.

15. The Process Control regulator tune-up may now be started.

CHAPTER 5
OPERATION INSTRUCTIONS

5-1. SCOPE OF PROCEDURE

The following procedure instructs the operator in the "power-up" and "power-down" of a typical 77-mm SILCOMATIC® power converter. The procedure assumes that the converter was initially checked out and calibrated for normal operation. Where equipment differences exist at various installations, refer to specific instructions supplied with the requisition.

5-2. PREPARATION FOR START-UP

1. Check that the process control drive regulator is suicided.
2. Ensure that the AC CONTROL SWITCH SCA on the converter is set to the TRIP position to prevent accidental closure during start-up.
3. Verify that all boards and ribbon connectors are in the proper position and are properly seated.

NOTE

It is good practice to determine the type of maintenance activity or any equipment/control changes made in the equipment during the "down" time. This will probably identify the area that should be closely checked before returning to normal operation.

4. Verify that all enclosure doors are securely closed except for the control panel containing the SEM and power supply.

5-3. CONVERTER START-UP

1. Apply control power by closing circuit breaker CPS (Figure 4-2). All bridge fan motors should be running.
2. Check that the power supply (PS) circuit breaker (Figure 4-2) is closed and that the amber LED (adjacent to the circuit breaker) is illuminated.
3. Check that the IMOK lights on all cards in both the SEM and the FM (Figure 4-1) come on when the computer finishes self-test.
4. Check that switches SW1 and SW2 on the HFXA card (Figure 4-3) are in the center position. Check that the Thumb Wheel Switch on the HFXA card is in position O.

5. Check that the door-mounted RIL lamp (Figure 4-1) is on. Check that all of the red LEDs on the HFPB card (Figure 4-3) are off. Check that only LEDs A, B and F on the HFXA card are on.

6. If a terminal is connected to the RS232 port on the HFXB card, check that the message reads SELFTEST COMPLETE: COMPUTER OK - SYSTEM INITIALIZED. Ignore the bridge test fault message at this time.

7. Set the AC CONTROL SWITCH SCA (Figure 4-1) to CLOSE position.

8. Close the ac primary disconnect.

9. Verify that the three white bridge power phase lights on the ac entry panel (Figure 4-1) are on.

10. Check that the amber phase lock loop LED (PLL), located on NBIB card (Figure 4-3), is illuminated.

11. Close the dc circuit breaker (if supplied).

12. If a terminal is connected to the RS232 port on the HFXB card, enter a "B" into the terminal. This command makes the computer request Bridge Test only. If a terminal is not available, press switch SW2 on the HFXA card downward. This is the master reset switch and by toggling it, the computer will repeat both the self-test routine and the bridge test routine. The bridge test routine has to be repeated here since it did not successfully complete the test on initial power up because bridge power was not applied.

13. When the computer finishes bridge test, check that the door-mounted RIL lamp is off. Check that all red LEDs on both the HFPB and HFXA cards are off.

14. If a terminal is connected to the RS232 port on the HFXB card, check that the message reads BRIDGE TEST COMPLETE: BRIDGE O.K.

    SILCO READY

15. Close the control panel door to ensure environmental protection during system operation. The power converter is now ready to be operated from the dc drive controls.

16. The dc drive control will release override control of the converter. The red firing pulse LEDs on the HPTN cards (in the Firing Module, Figure 4-1b) should be illuminated. One or the other LED will be on, depending upon the direction of operation.

    The 77MM SILCOMATIC power converter is now operating.

5-4. CONVERTER SHUT-DOWN

NOTE

The shut-down procedure at various installations may vary, particularly with regard to the removal of primary ac power.

1. Power-down or suicide the dc drive control (and motor).
2. Trip the dc breaker (if so equipped).
3. Set AC CONTROL SWITCH SCA (Figure 4-1) to the TRIP position and lock out the ac primary disconnect.
4. Open the control panel door.
5. Open the control power circuit breaker CPS (Figure 4-2).
6. Close all enclosure doors to discourage unauthorized entry since the equipment contains high voltage circuitry.

5-5. PROCESS/SILCOMATIC EQUIPMENT PROTECTIVE CIRCUITRY

The 77-mm SILCOMATIC equipment protective circuitry may fault-stop or controlled-stop a drive as a result of abnormal drive operating conditions or as a result of the main ac power source being outside of allowable operating specifications. In these cases, the process control equipment will receive a SILCOMATIC Fault (SFLT) signal which indicates that the power converter has gone to its retard limit or tripped off line, and/or a Power Converter Annunciate (PCA) signal which indicates that a mis-operation may occur if the drive does not change operating conditions. Such a signal will normally result in a controlled-stop of the process. The process control equipment will normally indicate that a protective device has operated in the power converter as well as in the process control circuitry. The following paragraph contains a simple procedure for identifying the protective circuitry that has operated and the corrective action required to return to service provided there has been no circuit failures other than SCRs.

NOTE

It is recommended that operating conditions and the status of monitoring devices be entered in a log.

5-6. SILCOMATIC POWER CONVERTER FAULT DETERMINATION

1. The CONVERTER FAULT indicator light on the control panel door (Figure 4-1) should be on.

2. Check that all three bridge power phase lights on the ac entry panel door (Figure 4-1) are on.

3. Open the control panel door and observe the red LEDs on the HFPB and HFXA cards (Figure 4-3). The first fault that occurred will have a flashing LED.

WARNING

To replace blown fuses and/or cells detected in steps 2 and 3, above, will require power turn-off of the power converter.

4. Refer to Table 8-1 (or 8-2 if operating in the alternate operating mode) for the definition of the LEDs.

5. If there was a terminal connected to the RS232 port on the HFXB card when the fault occurred, read the fault message describing the problem.

6. If there was no terminal connected to the RS232 port on the HFXB card when the fault occurred, connect the terminal to the port and enter an "S". This will dump the message queue to the terminal. The last message that the computer outputs represents the last fault that occurred.

7. If a shorted SCR is suspected, repeat the bridge test by entering a "B" into the terminal. The computer will identify any shorted SCR.

8. After the fault has been cleared, the hardware can be reset by toggling the RESET switch (SW2) on the NBIB card (Figure 4-3).

NOTE

The process control has to be suicided before the 77-mm Silcomatic circuitry can be reset.

9. Refer to Chapter 8 for additional troubleshooting information.

CHAPTER 6
SUBSYSTEMS DESCRIPTION

6-1. INTRODUCTION

This section provides a substantive description of each converter subsystem, including a discussion of major signal flow. A block diagram of the subsystems appears in Figure 2-2.

6-2. MAIN POWER DISCONNECT

Primary protection for the power converter is normally provided by remote, metal-clad type switchgear. Power converter isolation is often provided by a load break disconnect. If an isolation switch is used, fuses in the power disconnect limit the available fault current. An ac permissive switch is provided at the converter for tripping and locking out the main ac breaker. However, the main power disconnect is external to the power converter and is not supplied with the converter.

6-3. RECTIFIER TRANSFORMER

An isolating type transformer provides proper circuit and regulation characteristics and limits available fault currents to within acceptable limits. A thermal switch in each winding provides overtemperature detection. Also, a low oil level switch is provided in liquid-filled types.

6-4. FILTER AND VOLTAGE SURGE SUPPRESSION

Line-to-Frame Filter Capacitors are used to suppress voltage spikes caused by switching surges. They provide a discharge path between the secondary windings of the transformer and ground.

Line-to-Line AC Filters prevent voltage oscillations which result from switching of the thyristors as commutation progresses from one phase leg to the next. This effect becomes more severe as phase retard increases to the point of maximum ripple (between 60 and 90 degrees retard, depending on actual inductance of the load circuit), where the switching change of voltage approaches peak line-to-line ac volts ($2E_{L-L}$). Without these filters, overshoot would not only stress the inverse blocking capability of the cells, but ringing could occur sufficiently to switch off the cells after firing.
DC Filters, calculated for the rated load and inductance values in the dc bus loop, are used for damping purposes within the dc system and to aid commutation when the dc current is discontinuous between successive conduction periods.

Surge Suppressors, connected line-to-line, provide additional protection against transients caused by chopping excitation current in the converter transformer. Their function is to clip line-to-line surges at a value between the peak voltage level of the circuit and the peak inverse voltage rating of the thyristors. If the spillover conduction point of the suppressor is

reached, the impedance of the device changes instantaneously from a high value to a very low value. It then remains at the low value until the ac line-to-line voltages go negative, resetting the device.

6-5. CURRENT TRANSFORMERS

Main ac current transformers (CTs) are located on the ac bus for regulation and protective sensing. The secondaries of these transformers feed the primary winding of the auxiliary transformers. The secondary windings of the auxiliary transformers feed the burden resistors mounted on the Firing Module card (NATF). The signal obtained from the burden resistor is directly proportional to the current in the dc bus. This signal is used for fault monitoring and can be used for regulator feedback.

6-6. SIGNAL ELECTRONICS MODULE (SEM)

The DS3820SSP 77-mm SILCOMATIC Plus Signal Electronics Module (SEM) provides both 6-pulse gating and protective functions for the Silicon Controlled Rectifier (SCR) bridge, and can serve up to four parallel bridges. The gating pulses are produced from a voltage reference input provided by a process control regulator, which is external to the SILCOMATIC Plus system.

The DS3820SSP module (SSP) Form A consists of twelve microprocessor-based boards; SSP module Form B adds the DS3800NVIA Ground Detector Board. These boards are functionally grouped as follows:

Microcomputer Resources Functions	Generalized I/O	Custom System
DS3800HFPB Functional Processor Converter	DS3800NISB Signal Isolator	DS3800NCCB
DS3800HFXA Function Expander	DS3800NBIB Bridge Interface	Coordination
DS3800DFXA Daughter Board	DS3800NBIC Bridge Interface	DS3800DCCB
Daughter		
DS3800HFXB Function Expander	DS3800NBID Bridge Interface	Board
DS3800DFXB Daughter Board	DS3800NVIA Ground Detector	

The microcomputer resources are based on the 8085 microprocessor, and contain product software in Programmable Read-Only Memory (PROM) chips. Generalized I/O provides interface signal buffering and conditioning between the internal microprocessor and external hardware. This category includes gate pulse buffers, feedback conditioning, phase-lock loop, command signal isolation, and sensors for ambient temperature, current imbalance, and cell state. Custom system functions contain hardware unique to the 77-mm SILCOMATIC Plus system, including all variable analog scaling adjustments and control functions independent of the processor (such as providing an interface between 12-pulse and 24-pulse bridge sections).

The SEM component boards function together to provide bridge control and protection, as described in the following paragraphs. For more detailed information about a particular board, refer to the individual board's "Description of Operation and Application Data" (section 2), which is included in Tab 3.

6-6.1. DS3800HFPB Functional Processor

The DS3800HFPB card (HFPB) serves as the SSP module's internal processor. The HFPB contains an 8085 microprocessor, two PROM chips, five input/output (I/O) ports, an 8253 clock, a digital to analog (D/A) converter, and a six-channel, eight-bit analog to digital (A/D) converter. The processor is programmed to provide system-dependent software functions. The HFPB reads all SSP input signals and acts upon them according to software specifications. Additionally, the processor generates all reset and enable signals.

6-6.2. DS3800HFXA/DS3800DFXA Function Expander

The DS3800HFXA card (HFXA) with the DFXA daughter board expands the functions of the HFPB processor. The HFXA contains three input ports, four output ports, two PROM sockets, and an eight-channel, ten-bit A/D converter. The daughterboard contains two spring-return switches and one thumbwheel switch. The thumbwheel switch selects different outputs for display on the eight HFXA LEDs and eight HFPB LEDs, which indicate fault status. (Refer to Tables B-3 and B-4 for the assigned function of each LED.)

6-6.3. DS3800HFXB/DFXB Function Expander

The DS3800HFXB card (HFXB) with the DFXB daughter board further expands the functions of the HFPB processor. The HFXB contains two eight-bit D/A converters, two PROM sockets, one decoder PROM, and one four-bit output port. An RS-232 port allows communication with the HFPB processor through either a handheld or a standard RS-232 compatible terminal. The DFXB daughterboard contains 256 bytes of non-volitile read/write memory used to store requisition variable software.

6-6.4. DS3800NISB Bridge Interface

The two DS3800NISB cards (NISB) are analog and digital receiver/driver boards that provide an interface between the SILCOMATIC Plus system and the SSP's processor. Each board contains a relay driver, three inverters, four status latches, and the circuits described below.

6-6.4.1 VOLTAGE REFERENCE. An analog buffer circuit on the NISB contains a voltage isolator operational amplifier (op-amp) that receives a voltage reference signal from the SILCOMATIC Plus system's process control. This reference signal is a voltage proportional to the desired output voltage from the bridge: it varies from -10 V dc at rated reverse output to +10 V dc at rated forward output.

6-6.4.2. DETECTOR CIRCUITS. Four signal isolation circuits on each board monitor system components via normally-open relay contacts in series with a 120 V ac source. These circuits generate the following output signals:

- DC Breaker Closed (DCBC) -- not latched, indicates the state of the DC breaker. LED B on the front board edge indicates status.

- o Transformer Overtemperature (TOT) -- latched, indicates that the transformer has exceeded its rated temperature. LED C on the front board edge indicates status.

- o Fan Loss (FLOS) -- latched, indicates failure of a fan located in the cooling airflow of the SILCOMATIC Plus bridge. LED A on the front board edge indicates status.

- o Spare

6-6.4.3. LINE RECEIVERS. Four line receivers on each board provide input signals for system communication with the SSP as follows:

- o Suicide -- from the process control informs the processor to go to retard limit. This input can occur during a drive fault condition or during a normal stop. LED E on the front board edge indicates status.

- o Alternate Converter Okay -- indicates status of the alternate converter in a 12-pulse configuration. LED H on the front board edge indicates status.

- o Remote Forward -- is used to coordinate reversal control of 12-pulse systems. LED F on the front board edge indicates status.

- o DC Breaker High Speed Trip Ready -- indicates high speed trip control circuits are ready.

- o Enable -- from the Process Control Regulator. Allows gating of bridge cells to produce dc output. (on the NISB card in SEM slot 1A)

6-6.4.4. LINE DRIVERS. Four line drivers on each board provide the following SSP outputs, transmitted from the processor as follows:

- o Silco Fault -- to the process control, indicates that a fault has occurred in the SILCOMATIC Plus converter, and the power converter has tripped off-line or gone to retard limit (minimum output).

- o Power Converter Annunciate -- to the process control, can be a duplicate of the Silco Fault signal or several annunciate signals. Refer to the elementary provided with your system to determine how the signal is used.

- o Thermal Overload Forward -- to the process control, provides a fault indication. (Options are available for this signal. Refer to the elementary provided with your system to determine its use.)

- o This signal can be used either as "Thermal Overload Reverse" or as Alternate Converter Okay" for a 12-pulse configuration. Refer to the SILCOMATIC Plus converter elementary to determine which option is selected.

When using SSP Form B, the following signals on the NISB in SEM slot 1A may be used:

- o 0PLRST -- a logic 0 true output signal to the digital regulator to provide synchronization between regulator functions and cell firing events.
- o 1REFO -- a logic 1 true output signal to the digital regulator to serve as a reference for its local PLL circuit when no line voltage reference signal is available.
- o 1GNFLT -- a logic 1 true output signal generated by an NVIA board to indicate detection of a ground fault.

6-6.5. DS3800NBIB Bridge Interface

6-6.5.1. PHASE-LOCK LOOP. A phase-lock loop function on the NBIB board maintains proper firing order and timing, as well as noise isolation, by synchronizing bridge thyristor gate pulses with respect to the AC line. The loop's differential amplifier compares attenuated phase 1 and phase 3 line voltage input signals (buffered on the NBIC) to produce a V1-V3 signal. A reconstruction circuit filters out commutation notches and noise in the signal. The reconstructed V1-V3 signal is then integrated and amplified into a sine wave, which is phase shifted 90° from the original V1-V3 signal. The shifted V1-V3 signal is sent to a zero-crossing detector, which produces a logic 1 at the 90° point and a logic 0 at the 270° point of the input signal. The resulting square wave output signal is then used as the reference input to the phase-lock loop.

The output of the phase-lock loop is applied to an angle counter. Feedback detection provides a signal that changes to a logic 1 when the counter reaches 90°, and to a logic 0 when the counter reaches 270° (the same as the reference signal). The phase-lock loop control monitors the transitions of both the reference and the feedback signal. It forces a higher voltage output from the voltage controlled oscillator of the loop if the reference signal transition precedes the feedback signal transition. Likewise, if the feedback transition precedes the reference transition, the phase-lock loop control forces a lower voltage output from the oscillator. As a result, this integrating type control locks the reference and the feedback signals at the same phase.

A 9-bit digital comparator receives both the 9-bit counter outputs from the phase-lock loop, and a 9-bit signal from the processor that represents the calculated firing angle of the next SCR firing. When the phase-lock loop's output equals the specified firing angle, the comparator transmits an interrupt to the processor, which then initiates the cell firing.

6-6.5.2. TEST OSCILLATOR. The processor provides a test clock which can be used in testing bridge and firing circuit operation prior to applying bridge power. The NCCB board selects the test clock to provide a replacement signal for the phase-lock loop's V1-V3 line synchronizing signal, which is derived from the incoming bridge power. This replacement signal then allows the firing circuit to operate and gate pulses to be applied to the SCRs. The test oscillator is sleected by placing both switches SW1 and SW2 on the HFXA card in their upper positions.

6-6.5.3. INDICATING LEDS. The NBIB contains six indicating LEDs that monitor outputs of the SCR cell state sensors (see Table 1). Zero volts across an SCR (as with a shorted SCR) turns off the corresponding LED. These LEDs have meaning only when gate pulses are not being applied.

Table 1
CELL STATE INDICATING LEDS

SCR	LED
1F/4R	C
3F/6R	D
5F/2R	E
4F/1R	F
6F/3R	G
2F/5R	H

6-6.6. DS3800NBID Bridge Interface

6-6.6.1. PARALLEL BRIDGE CURRENT SHARING. Current sharing between parallel bridges is forced by current transformers on each bridge's incoming ac bus. (The SSP serves up to four parallel bridges). The NBID contains leg current sensors that monitor signals from these current balance transformers. The processor reads the sensor outputs to detect inequality (current imbalance) between the bridges, which can be caused by gate pulse loss or cell failure.

6-6.6.2. AMBIENT TEMPERATURE AND AIRFLOW RATE. The SSP provides ambient temperature sensing for compensation of the SCR thermal model, and signals airflow loss in the power converter bridge's cooling system. A sensor board mounted within the bridge transmits representative analog signals for ambient temperature to the NBID. There, an op-amp amplifies the ambient temperature input to provide a voltage representing ambient temperature. The NBID transmits the temperature voltage to the HFXA's A/D converter, which sends the
digital value of each to the HFPB processor. The processor compensates the SCR thermal model in accordance with the ambient temperature. Fan-mounted centrifugal switches provide a loss of airflow signal; if the processor detects a loss of airflow, it shuts down the SILCOMATIC Plus converter.

6-6.7. DS3800NCCB/DS3800DCCB Converter Coordination

The NCCB and daughterboard DCCB provide various converter coordination functions, as described below. The DCCB contains scaling potentiometers for adjustments that depend on the rating of the individual drive. (Refer to the SILCOMATIC Plus elementary supplied with your equipment for adjustment requirements.)

6-6.7.1. CURRENT FEEDBACK. The NCCB receives a current feedback signal (a rectified ac current transformer signal) from the burden resistor in the firing module. Scaling potentiometers on the DCCB are adjusted to scale the current feedback so that 10V = maximum short-time overload. The voltage is transmitted to an A/D converter on the HFXA, which sends the digital value of the feedback signal to the HFPB processor. The circuit also provides an ammeter driver signal, adjusted by a potentiometer and transmitted to an external dc ammeter.

6-6.7.2. VOLTAGE FEEDBACK. A voltage feedback circuit on the NCCB receives attenuated voltage signals (a voltage proportional to the bridge voltage) from the bridge P bus and N bus. (The voltage feedback scaling potentiometer must be adjusted.) The circuit's voltage output is transmitted to an A/D converter on the HFXA, which sends the digital value of the feedback signal to the HFPB processor. The circuit also provides a voltmeter driver signal, adjusted by a potentiometer and transmitted to an external dc voltmeter.

6-6.7.3. CEMF. A CEMF (counter electro-motive force) signal is derived from the attenuated voltage signals from the bridge P bus and N bus (the scaling must be adjusted per elementary specifications). The circuit's output voltage is transmitted to an A/D converter on the HFXA board, which sends the digital value of the feedback signal to the HFPB board. This signal is used in reversal control. The CEMF signal must be scaled the same as the voltage feedback signal (0 to +10 V dc for 0 to $\pm$ V_{out})

6-6.7.4. FEEDBACK AVERAGING. For SILCOMATIC Plus converters that operate in the 12-pulse and 24-pulse modes, the NCCB board contains feedback averaging circuits for both current and voltage feedback. The averaging circuit receives feedback signals from each converter, averages the signals, then transmits the averaged signal to the processor for current balance regulation. The averaged feedback signals may also serve as feedback signal outputs for the process control.

6-6.7.5. SUPPRESSION. The NCCB contains an instantaneous overcurrent circuit (IOC) which compares current feedback with an adjustable set point to detect high current level. If current level is high, the IOC initiates a protective action in which the firing circuit phases back to inverter limit until current drops to zero, then suppresses gate pulses. The high speed trip circuit (see below) trips the dc breaker, if present. An IOC potentiometer on the DCCB board establishes the IOC trip point.

6-6.7.6. DISCONTINUOUS CURRENT DETECTOR. The NCCB contains a circuit that analyzes the voltage feedback waveform to detect discontinuous current produced by the power converter. If discontinuous current is detected, the firing circuit adjusts the firing angle to linearize the firing circuit gain. This function provides equal gain for both continuous and discontinuous current.

6-6.7.7. HIGH SPEED TRIP CIRCUIT. The dc breaker high speed trip circuit trips the dc breaker upon loss of control power, dc fault detection, or through IOC initiation. When the breaker is tripped, current drops to zero in one cycle.

6-6.7.8. SHUNT TRIP CIRCUIT. The dc breaker shunt trip circuit is a 28-volt relay driver that controls a relay which trips the dc breaker. When the breaker is tripped, current drops to zero in 3 cycles. (This shunt trip function is a back-up to the high speed trip function.) The normal state of this relay driver is energized, and it drops out to initiate the shunt trip.

6-6.7.9. AC BREAKER TRIP CIRCUIT. The ac breaker trip circuit is a 28-volt relay driver that controls a relay which trips the ac breaker. This breaker trip is initiated by ac fault detection, IOC detection, or by switching the SILCOMATIC Plus converter into the system test mode.

6-6.7.10. TEST OSCILLATOR SELECTION. The NCCB contains a circuit that decodes signals for selection of the SSP's test oscillator located on the NBIB board. This test oscillator function is used for testing the firing circuit and bridge operation before applying bridge power. It is selected by placing switches SW1 and SW2 (HFXA) in their upper positions.

6-6.8. DS3800NBIC Bridge Interface

6-6.8.1. GATE PULSE LOGIC. The NBIC contains gate pulse logic that conditions the gate pulses and transmits them to the firing module. The actual cell firing signals are generated by the processor, then amplified by Darlington transistors on the NBIC. The circuit contains twelve gate pulse amplifiers, enabling the SSP to drive a full reversing 6-pulse bridge.

6-6.8.2. CELL VOLTAGE SENSORS. The NBIC contains voltage sensors that detect voltages across the cells as a check for cell firing. When a cell fires, the voltage across it drops to nearly zero. The NBIC measures each cell's voltage drop, then informs the processor and turns off the related indicating LED (located on the NBID board) if the cell fails to fire.

6-6.8.3. PHASE AND SEQUENCE SENSOR. The voltage sensors also provide phase and sequence loss detection. The NBIC creates V1-V2 and V1-V3 signals from the voltage sensors' outputs. The processor uses these circuit outputs to verify proper input power and to prevent drive operation if it detects a wrong phase or sequence.

6-6.9. DS3800NVIA Voltage Isolator

The NVIA board, used only in Form B, monitors the algebraic sum of the positive motor terminal voltage and the negative motor terminal voltage to detect an impedance imbalance. An imbalance indicates the presence of a ground fault on the motor.

The NVIA provides the following output signals:

- o IGND --a logic 1 true output that indicates a ground fault detection. The signal feeds the processor port B (bit 6) and a line driver.
- o VGND -- an analog output that shows the relative magnitude of the ground fault. VGND is provided on the NVIA's connector JF to drive a panel meter.

6-6.10. Other Features

6-6.10.1. MODULE SELF-TEST. The processor performs hardware self-tests on the SSP's PROMs and RAMs at power-up. If any test fails, the processor halts and then sends an appropriate message to the terminal via the RS-232 port. Next, the processor conducts bridge tests that check phase rotation, phase-lock loop operation, and the bridge for open and shorted cells. If any of these tests fails, the processor halts, then sends an appropriate message to the terminal. (For a list of the different self-tests and bridge tests with their corresponding messages, refer to Appendix A.)

6-6.10.2. ON-LINE DIAGNOSTICS. For troubleshooting, the HFXA's and HFPB's fault status LEDs can be reassigned to certain data in memory via the DFXA's thumbwheel switch and 2 toggle switches. These data assignments are listed in Tables 8-1 and 8-2 in Chapter 8.

Two software controlled analog outputs (the HFXB board's TP1 and TP2) provide real-time viewing of several selected software variables in memory. Another software controlled analog output (the HFPB's TP2) provides a "trace" waveform that can be displayed on an oscilloscope to show different levels of processing activity. These analog outputs, and appropriate scaling factors, are indicated in Table 8-3.

Additionally, the processor maintains a 1024-byte circular list fault finder with variable specification that can be modified via the RS-232 port. The data contained in the circular list can be accessed by either the RS-232 port or the HFXB's analog output ports, TP1 and TP2.

6-7. SIGNAL ELECTRONICS POWER SUPPLY

This base-mounted power supply accepts 110 V ac as an input and develops +15, -15, +5 and +28 volts for use by the Signal Electronics Module.

Circuits included are:

1. Incoming power circuit breaker
2. Current limit protection on all buses
3. Remote sensing on +5 volt bus
4. Potentiometer adjustment of all buses except +28 V
5. Berg jumper adjustment for +28 V
6. Power supply status indication
7. Logic output to the Signal Electronics Module. Low voltage on any bus is detected.
8. Undervoltage and overvoltage shutdown

6-8. FIRING MODULE (FM)

The DS3820FMAA module (FMAA) consists of a backplane board DS3800NATF and six firing boards DS3800HPTN. The firing boards are plugged directly into the NATF and are supported by card guides as part of the FMAA structure.

The FMAA is a multipurpose module intended to process the following signals (refer to Figure 6-1):

1. Gate pulses to the power module thyristors.
2. Current feedback from power module current transformers (CTs).
3. Analog voltages from the power module, in conjunction with attenuator board DS3800NATA (located elsewhere).
4. Temperature signals from board DS3800NTSC located in the power converter module.
5. Current balance transformer voltages.

6-8.1. Thyristor Gate Pulses

The six HPTN boards receive logic commands to fire thyristors from the control SEM. Each HPTN board has two pulse channels, for forward and for reverse gating. The boards are located from left to right in the firing sequence 1, 3, 5, 4, 6, and 2.

The command gate signals are received from the control SEM via ribbon cables into connector JA on NATF. The signals are distributed to the six HPTN boards by NATF circuit runs. The HPTN boards convert the logic command signals into current pulses that are sent to gate pulse transformer primaries. The output connections are located in plugs JC (forward) and JB (reverse) on the edge of each HPTN board. The gate pulse transformers are located in the power converter module, inside the snubber and gating module near the power thyristors.

Indicator lamps are provided on the edge of the HPTN boards. The red lamps indicate when gate pulses are present. The amber lamps (IMOK) are normally on; If any IMOK lamp is off while the P70 supply is energized, it indicates a malfunction.

The IMOK lamp is connected to an undervoltage detector circuit in the HPTN. In the event of a malfunction, both the IMOK lamp goes off (local indication) and a logic 1 is sent from the card. TP11 on the HPTN card monitors the logic signal. The logic signals of all six HPTN boards are combined in an "or" logic circuit on the NATF board. The output of the "or" circuit is sent via plug JB to the control SEM for monitoring. The "or" logic signal can be monitored at TP1 on the NATF card.

Each HPTN board is provided with a protective fuse located on the board, towards the back of the module. This fuse will open if the HPTN is overloaded.

6-8.2. Current Feedback

Two current sense transformers (CTs) are used in the power module. The CT lines are connected to the FMAA module by plug JQ. The current is full-wave rectified by a diode bridge. The signal is sent to the control SEM via plug JA.

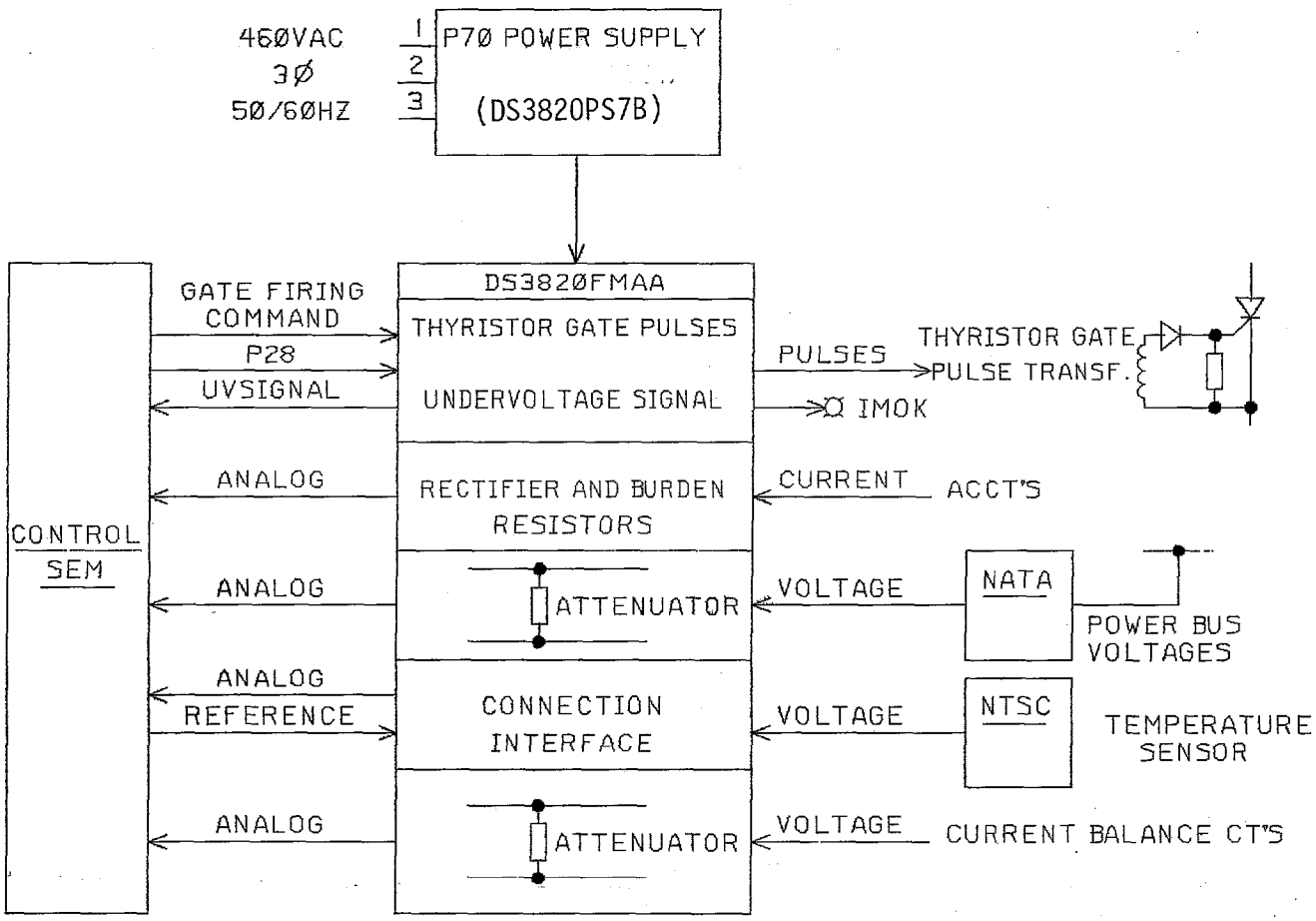


Figure 6-1. Firing Module Block Diagram

6-8.3. Attenuator Voltages

A board DS3800NATA is used in the power converter module to sense five power circuit voltages: the three ac line voltages and the positive and negative output bus voltages. These signals are attenuated by series resistor strings in the NATA board and by resistors in the FMAA module. The connection of the NATA lines to the FMAA module is done by plug JK (10-line ribbon cable). Resistors R26 through R36 on the NATF board complete the attenuator network. Berg jumpers BJ2 through BJ6 on the NATF card select the 12.1 k ohm resistor. Jumpers on the NATA card control the attenuation ratio and must be set in accordance with sheet AB of the elementary diagram. The attenuated signals are sent to the control SEM via plug JA.

6-8.4. Temperature Sensors

A maximum of four temperature sensor boards DS3800NTSC may be connected to the FMAA through plugs JR, JS, JT, and JU. The signals are sent to the control SEM via plug JC. Each sensor is independent and receives its own reference from the control SEM. These sensors, located in the Thyristor Power Module (described below), are used to monitor ambient temperature and air flow in the power conversion module.

6-8.5. Current Balance Transformers

If more than one converter bridge is used in the power module, current balance transformers are required to provide current sharing between the converter bridges. A maximum of four bridges in parallel may be used. The current balance transformer secondaries are monitored by the control SEM to determine if an unbalanced condition exists between the bridge currents. The signals from the transformers are brought to the FMAA module through plugs JL, JM, and JN for power phases 1, 2, and 3, respectively. Signal attenuation is achieved by resistors R11 through R22 located on the NATF board. These resistors provide attenuation in conjunction with resistors that are mounted on the current balance transformers. Capacitors C7 through C18 are used for signal filtering purposes. The attenuated signals are all available at test points TP2 and TP4 located on the HPTN boards.

6-8.6. 70 V Power Supply

A 70 V dc supply is connected to the FMAA module by plug JP. It is a regulated supply operating from the 3-phase, 460 V ac, 60/50 Hz control power supplied by the user. This supply provides power to the gate driver amplifiers on the six HPTN boards. There are also two ground stab terminals GNDA and GNDB to connect the transformer electrostatic shields to the frame ground.

6-9. THYRISTOR POWER MODULE

The function of converting ac power to adjustable dc power is performed by the Thyristor Power Module. This is made possible by the characteristic of the thyristor to block voltage in both directions while the degree of current flow in one direction is controlled by gating the thyristor at adjustable times during the applied ac waveform. Non-reversing and reversing configurations are available; simplified schematic diagrams of these are shown in Figures 6-2 and 6-3, respectively.

The module comprises a reversing or non-reversing full-wave thyristor bridge using 77-mm air-cooled thyristors. The items comprising the module are as follows:

Thyristor Module
Snubber Circuits
Leg Reactors
Current Balance Transformers
Pulse Transformers
Thermal Sensors
Leg Fuses (3- and 4-bridge drives only)

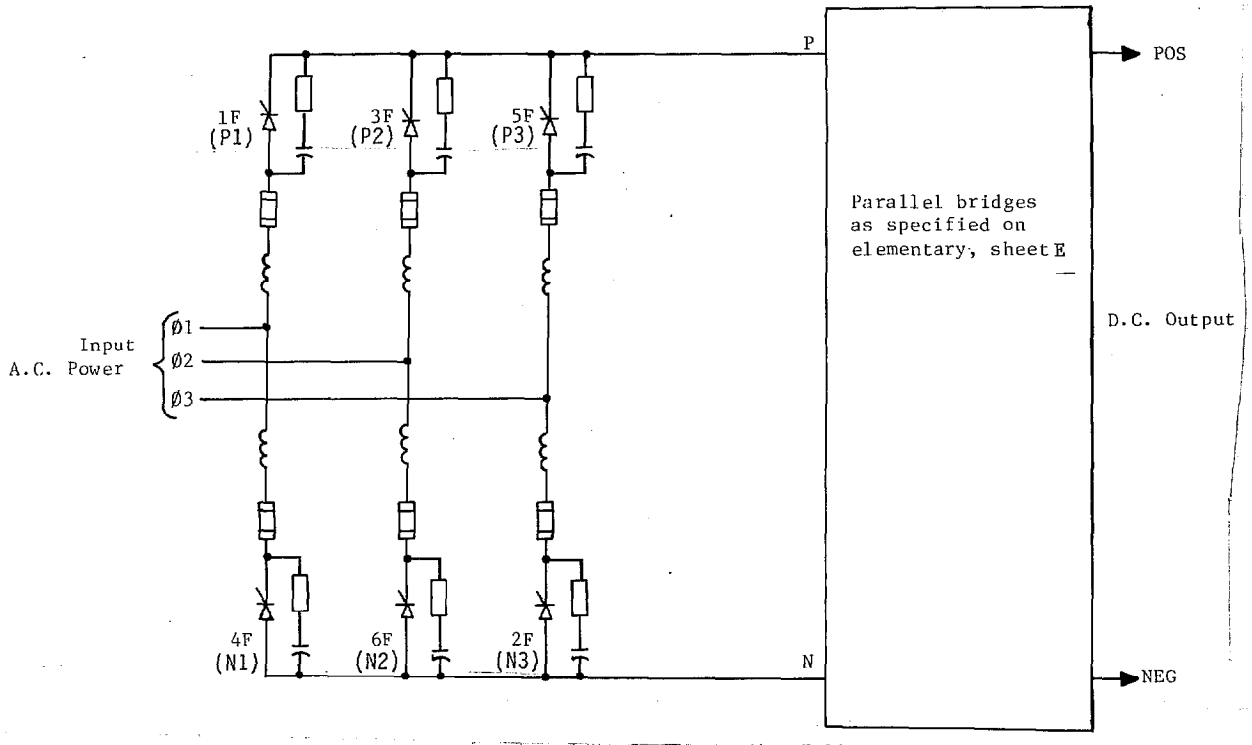


Figure 6-2. Non-Reversing Bridge Configuration

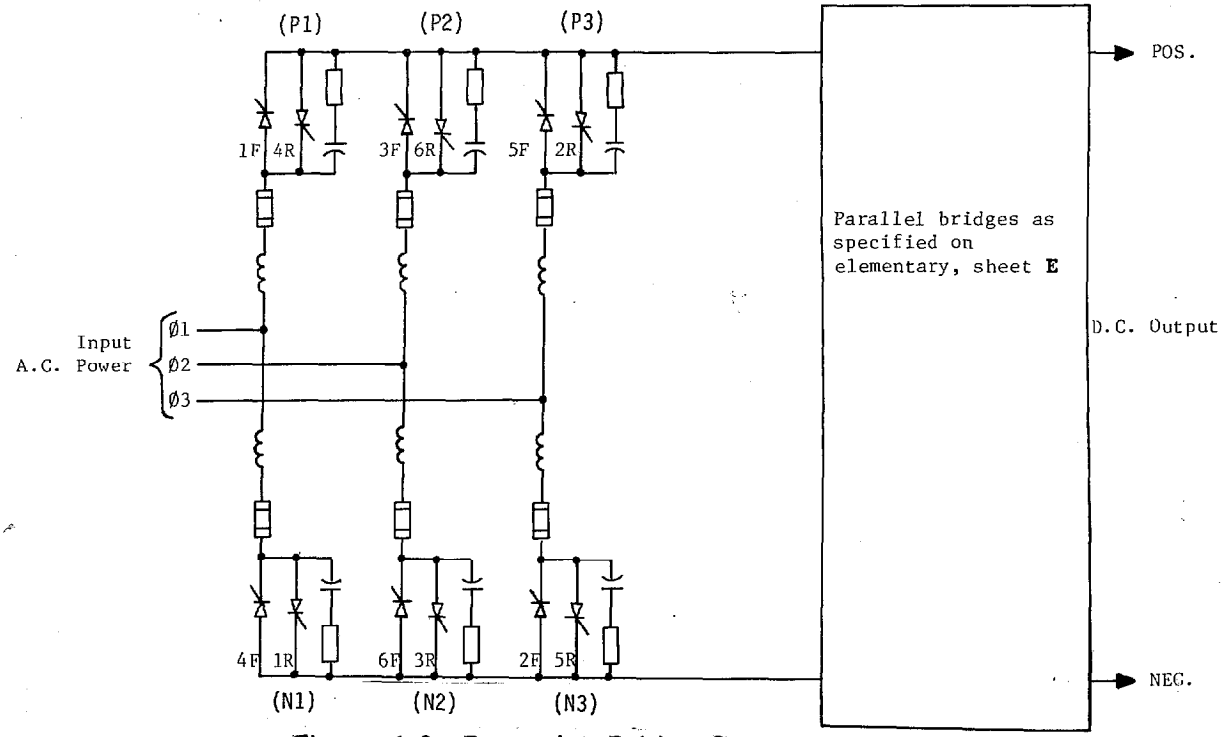


Figure 6-3. Reversing Bridge Configuration

6-9.1. Thyristor Module

The thyristor module consists of one or two thyristors assembled as a stack with cooling fins for forced air cooling. When the two-thyristor stack is used, the bridge is capable of full reversing operation. The thyristors are assembled so that they are in reverse parallel.

6-9.2. Snubber Circuits

Each stack requires a snubber circuit connected across the thyristor terminals, from anode to cathode. The snubber circuit consists of one 0.82 uF capacitor in series with a resistance of 27 ohms, made up of six 40 ohm elements in series/parallel. The same snubber circuit is used for the reversing or non-reversing module.

6-9.3. Leg Reactor

Two ferrite inductor assemblies are connected in series with the thyristors. These inductors limit the rate of current increase thru the thyristor during turn-on.

6-9.4. Current Balance Transformers

Current balance transformers (CTs), omitted from single bridge power converters, are used to achieve current balance when bridges are operated in parallel. Each bridge has a current balance transformer installed in each phase of the ac input power. All transformer secondary windings of the same phase (one per bridge) are interconnected so that a current imbalance among the parallel-connected bridges produces a correction voltage that establishes (or restores) a balanced current distribution.

The CT secondary voltages are also monitored by the control system and an alarm annunciates an unbalanced condition. A resistive divider provides attenuated signals to the control SEM. Overvoltage protectors (MOVs) are used across the divider to protect control circuits.

6-9.5. Pulse Transformers

The power thyristors are gated with pulse trains coupled through the pulse transformers to the cell gates. Pulse commands are generated in the control SEM, amplified by the FMAA module, and sent to the pulse transformers.

In drives with multiple bridges in parallel, the pulse transformers for each phase are connected with their primaries in series. This achieves simultaneous gating of the parallel bridges, providing nearly uniform current division with less compensation by the current balance CTs.

The pulse transformers also provide electrical isolation between the power bus and the control circuit. An electrostatic shield in the transformer minimizes common mode noise coupling between the power module and the control. The electrostatic shield should always be grounded to the module frame.

6-9.6. Thermal Sensors

Each power converter bridge module includes a thermal sensor card DS3800NTSC. The card is mounted in the module, near the cooling air stream, and it monitors the temperature and velocity of the incoming cooling air. It is connected to the control SEM via the FMAA module. Should the incoming air temperature exceed the equipment rating, it will be detected by the control SEM and an alarm will result.

6-10. DC CIRCUIT BREAKER

In Class I protection drives, a high-speed-trip dc breaker is provided. A POWER BREAK® breaker is used when the rated dc current is 4,000 amperes or less. When rated current is greater than 4000 amperes,an ITE breaker is included in the converter-to-motor dc loop to interrupt ac and dc shoot-through types of faults. When a fault condition is sensed by the remote fault detection circuitry which continuously monitors the drive, the breaker is opened by the high speed trip coils. Fault current limiting begins in 5 to 8 milliseconds. The breaker can also be opened by turning the dc breaker closing switch (SCD, on panel door) to the TRIP position. In this case, the current limiting will begin in 40 to 55 milliseconds. The breaker is fed 230 volts ac to operate the closing motor, relaying, and lights. The breaker is closed by turning SCD to the CLOSE position. This operates the closing motor.

CAUTION

A 5-MINUTE REST PERIOD SHOULD BE ALLOWED BETWEEN CLOSING OPERATIONS OR THE CLOSING MOTOR MAY BE DAMAGED DUE TO OVERLOAD.

6-11. RIPPLE REACTOR

A remotely mounted ripple reactor (if used) is added in the dc loop to reduce, to within motor commutation and heating limits, the ripple produced by the power converter. The reactor is normally selected to limit the RMS sixth harmonic current at rated voltage and current output to 2% of the rated dc motor current.

CHAPTER 7
PREVENTIVE MAINTENANCE

7-1. INSPECTION ROUTINE

A periodic routine of cleaning and inspection should be established. Such cleaning is necessary to prevent faulty operation of a device due to accumulation of dust and corrosion. Inspections should be more frequent if demands on the equipment are exacting. Under normal conditions, the protective devices do not operate; therefore, these devices need careful inspection. If the equipment is in operation when being inspected and a short interruption of service can be permitted, the manual tripping of a protective device will afford an opportunity to observe the manner in which the equipment is functioning.

If, at any time, a device fails, the cause of the failure should be identified and corrected before any remedy is applied and, if necessary, the part adjusted or replaced.

NOTE

When ordering parts, give a detailed description of the part and complete nameplate data on the device.

For specific maintenance information on the various devices, refer to the particular device instruction book.



TO AVOID PERSONAL INJURY, AUTOMATICALLY CONTROLLED EQUIPMENT MUST NOT BE STARTED BY HAND UNTIL THE CAUSE OF THE FAILURE TO START AUTOMATICALLY IS DETERMINED AND CORRECTED.

7-2. RECORDS AND LOG BOOK

It is recommended that a regular inspection and maintenance program be employed and that a log book be kept. This procedure will help detect malfunctioning equipment and determine the cause of the trouble. It is suggested that a form be created and filled out at each regular inspection or when trouble is experienced. The form should contain the following data:

Normal Operation:

Operator
Date and time
Converter dc volts
Converter dc amps
Maintenance performed
Comments

Abnormal Operation - Additional Observations:

Alarm lamps
Tripped breakers
Terminal Messages

Circumstances at time of trouble, such as:

Start up
Shutting down
Load
Best operation
Disturbances
Abnormal operations elsewhere but still associated with the converter.

7-3. DEVICE MAINTENANCE

Since the power conversion equipment contains mostly static devices, very little maintenance is required other than to occasionally check that connectors, boards, wire terminal boards, and power bus connections are secured properly. Since the very few relays used are of the sealed type, they require practically no maintenance.

The devices which require some degree of maintenance are the dc circuit breakers (if used), the cooling fans, and heat sink assemblies. Specifically, the recommendations are as follows:

1. Fans: Since the fan motors are totally enclosed, outboard fan-cooled and contain permanently lubricated, sealed bearings, they require essentially no maintenance. However, they should be checked occasionally to ensure that fan blades and motor mounting brackets are tight. Under exceptionally dirty conditions, the motor outboard fan assembly should be inspected for restricted air passage-ways.

2. DC Circuit Breaker: Two dc circuit breaker designs are available, depending upon application:

   A. POWER BREAK circuit breaker
   B. ITE FBK-H circuit breaker

   The frequency and degree of maintenance required for these breakers is proportional to the number of operations and the amplitude of the load current interrupted. For specific maintenance requirements and instructions, refer to the following publications (included in Tab 4):

(1) POWER BREAK circuit breaker - refer to GEH-4355 and GEH-4382.

(2) ITE FBK-H circuit breaker - refer to IB 16.4.1.7-1C.

3. SCR Heat Sink Assemblies: Where extremely dirty conditions exist, there may be sufficient accumulation of foreign matter to restrict the air flow across the heat sink assemblies. Prior to performing any maintenance in the SCR power bridge, ensure that ac bridge power is locked out. To remove dirt from the heat sinks, they should be cleaned while mounted in place with a strong industrial vacuum cleaner using a brush attachment.

4. SCR Cell Stack Replacement: Refer to Sheets 1AA-1AD of GE drawing no. DS3826SS (included in Tab 3).

CHAPTER 8
CORRECTIVE MAINTENANCE

8-1. INTRODUCTION

This chapter describes several procedures to facilitate fault isolation in the 77-mm power converter. It begins with a check of the control power supplies, then proceeds to the programmed system diagnostic tests. Additional diagnostic "tools" are described, such as the circular list, software test points (D/A converters), and various test vectors.

8-2. REFERENCE DOCUMENTS

A thorough understanding of the principles of operation is the basis of effective fault isolation. This understanding can be achieved using the following reference data:

1. System description provided in Chapter 6.
2. Printed wiring board and module descriptions, including elementary diagrams, included in GEK-83800A.
3. System elementary diagram, supplied with each power converter.
4. Sequencing Logic Diagram, included in GEK-83800A.
5. Tabular data in Appendix A.

Chapter 6 and the printed wiring board/module descriptions included in GEK-83800A provide detailed circuit descriptions of the many printed wiring boards, modules, and subsystems in the power converter. The system elementary diagram, supplied with each power converter, also contains detailed one-line (functional) diagrams, and unique tune-up data. The Sequencing Logic Diagram, included in GEK-83800A, depicts schematically system programming. Tables in Appendix A of this book detail the various terminal message that may be generated during diagnostic routines.

8-3. CONTROL POWER SUPPLY CHECKS

1. Ensure phases 1, 2, and 3 on the incoming side of the Control Power Switch (CPS, Figure 4-2) are connected in the proper sequence.

2. Verify that the PSFF power supply circuit breaker (Figure 4-2) is open.

3. Close breaker CPS.

4. Using the DMM, measure the voltage between transformer T2 (Figure 4-2) terminals X1 and X2. The voltage must be 115Vac $\pm$ 10Vac.

5. Close circuit breaker PSFF.

6. Using the DMM, measure the voltage at Firing Module (FM) circuit card HPTN test point TP1 with respect to TP3. It must be between 66.5 and 84 volts.

7. Using the DMM, measure the following voltages. Adjust the indicated potentiometer (or Berg Jumper) to obtain the required indication:

Test Point	Reference Point	Voltage Range	Adjustment
+ 5 V	DCOM	4.95 - 5.05	R1 on NPPB
+ 15 V	ACOM	14.95 - 15.05	R6 on NPSK
- 15 V	ACOM	14.95 - 15.05	R7 on NPSK
+ 28 V	DCOM	27.0 - 29.0	BJ1 on NPSP

8-4. DIAGNOSTIC PROGRAMS

Each time the power converter is started (or reset), the internal computer executes a sequence of two diagnostic routines: computer self-test and bridge test. The computer self-test checks the PROM and RAM for defects; the bridge test checks line power phase rotation, power synchronization (phase lock), and for defective (open or shorted) SCR cells.

8-4.1 Computer Self-Test

1. Place the HFXA card switches SW1 & SW2 (Figure 4-3) in their center positions and the Thumb Wheel Switch (TWS) at 0.

2. Press and hold HFXA SW2 to start the computer self-test. Make sure that all Light Emitting Diodes (LEDs) on the HFXA and HFPB cards light while SW2 is held. Release SW2 to proceed with the self-test.

3. The computer self-test sequence is detailed in Appendix A. If any of the 18 tests fails, read the message sent to the terminal to identify the defective part. Alternatively, the HFXA and HFPB LEDs can be interpreted, using Appendix A, to determine the defective part.

4. When all tests are completed, a message will be sent to the terminal indicating the computer is OK.

5. When the computer finishes its self-test routine, it automatically starts the bridge test routine. Since there is no bridge power, the computer will fail bridge test and send a fault message to the terminal; ignore this message.

6. Check the IMOK lights on all cards. If any are not on, replace the card.

8-4.2 Bridge Test

1. Verify that the dc breaker is open.

2. Apply bridge power and observe that the PLL LED on the NBIB card comes on.

3. Press switch SW1 to begin the bridge test. At the conclusion of the test, the terminal will display either a fault message identifying the problem or the message BRIDGE TEST COMPLETE: BRIDGE OK.

8-5. DIAGNOSTIC AIDS

This section describes four diagnostic aids to fault isolation:

1. LED status indicators on the HFXA and HFPB cards in the Signal Electronic Module (SEM)
2. Software testpoints on the HFXB card
3. Output port test vectors to exercise the logic circuits
4. Circular list to monitor a maximum of four user-selectable variables.

8-5.1 LED Status Indicators

Light Emitting Diodes (LEDs) on the HFXA and HFPB cards monitor a variety of system functions. During the computer self-test, they indicate which of the 18 tests is being executed and, if the computer fails to satisfactorily complete its self-test, the test which failed. These definitions are included in Table A-1 of Appendix A.

During normal converter operation, these 16 LEDs display switch-selectable software variables or SCR cell status. The variable or device monitored is selected by the thumbwheel switch TWS (Figure 4-3) in conjunction with two toggle switches, SW1 and SW2 (Figure 4-3). LED definitions are presented in Tables 8-1 and 8-2. Table 8-1 defines the LEDs for each of the sixteen positions of the thumbwheel switch, while switches SW1 and SW2 are both in either the center or upper positions. Table 8-2 redefines the LEDs when switch SW1 is in its upper position while SW2 is in its center position. This mode, which can be selected while the drive is in operation, is called the Alternate Operating Mode. The only difference between this mode and the normal mode is the different LED definitions.

Table 8-1. LED Definitions (HFPB & HFXA Cards) -- Normal Operation & System Test Modes
(Switches SW1 & SW2 at Center Positions or SW1 & SW2 at Up Positions)

LED	Thumbwheel switch position															
	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HFPB card																
A	CURBAL.	CURBAL.	2DEG	UNSUIC	1DA	RSTPB	0EXCTL	AT60	TOC	RTDL	↑	↑	↑	↑	↑	X
B	MISPUL.	MISPUL.	4DEG	REMFWD	1AB.	PSFLT	0AVE2	FJT3	FLASH	SFLT	|	|	|	|	|	X
C	DIA.	DIA.	8DEG	HSTR	1BC	RTS	0AVE3	FJT2	LOCK	PCA	F	T	R	T	T	X
D	CIRC.	CIRC.	15DEG	ALTOK	1CD	FBLO	0AVE4	FJT1	RETST	STAT1	W	F	W	R	S	X
E	COMM.	COMM.	30DEG	UNLOC	2DA	-	0GOFWD	NOAIR	10K	STAT2	A	J	A	J	I	X
F	10C.	10C.	60DEG	CCBOK	2AB	-	0FFRST	RJT3	10TST	TRGED	T	U	T	U	N	X
G	PSUV.	PSUV.	120DG	P70UV	2BC	-	0CLRCT	RJT2	SYSTST	-	T	N	T	N	K	X
H	REVERR.	REVERR.	240DG	ERR	2CD	-	0REV1	RJT1	ALTOP	-	S	C	S	C	|	X
											↓	↓	↓	↓	↓	
HFXA card																
A	BRDGOK	↑	VP1	FANOK	SLCOK	3DA	0P1	ALPUL	NVROK	LAT60						X
B	DCBC	|	VP2	DCBC	CPCA	3AB	0P2	ALT	STOERR	LNOA1						X
C	10C	D	VP3	TRNOK	OSTA1	3BC	0P3	CLRAC	PLSERR	TOLF						X
D	TOLF	E	VN1	SPR1	OSTA2	3CD	0N1	RST	ALTLO	TOLR						X
E	TOLR	P	VN2	FANLOS	FLTL1	10C	0N2	RECAL	BRDGOK	FJTA						X
F	LUNLOC	T	VN3	LDCBC	TRIPA	V1V2	0N3	XX	DRSEL	RJTA						X
G	LAT60 [a]	H	TMORE	TOT	RLO	V1V3	0PLRST	XX	RUN	LUNLO						X
H	PSUV	↓	TLESS	LSPR1	DCSTA	1ZERO	0REVER	XX	STOPD	PSUV						X

a FBLO when NATF card current feedback plug is disconnected.

Table 8-2. LED Definitions (HFPB & HFXA Cards) -- Alternate Operating Mode
(Switches SW1 Up & SW2 at Center Positions)

LED	Thumbwheel switch position															
	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HFPB card																
A	CURBAL.	CURBAL.	P1	A	P1CELL	--	A						X			X
B	MISPUL.	MISPUL.	P2	L	P2CELL	--	I						X			X
C	DIA.	DIA.	P3	P	P3CELL	--	R	P	N				X		B	X
D	CIRC.	CIRC.	N1	H	N1CELL	--	F	1	1	1	C	1	X	1	A	X
E	COMM.	COMM.	N2	A	N2CELL	--	L	5	5	R	E	F	X	A	L	X
F	IOC.	IOC.	N3	F	N3CELL	--	O			E	M	B	X	V	E	X
G	PSUV.	PSUV.	1=R;0=F	L	REV	MSGS	W			F	F	K	X	E	R	X
H	REVERR.	REVERR.	FLTFLG	T	OPEN	CBCTF							X		R	X
HFXA card																
A	BRDGOK		PHFLT	C	NOPLL	NOPLL	A						X			X
B	DCBC		PHFLT	E	NOFR1	NOFR1	M						X			X
C	IOC	D	PHFLT	L	MPLLC	MPLLC	B		P				X			X
D	TOLF	E	DFFLG	S	REVPR	REVPR	T	P	2				X			X
E	TOLR	P	IFFLG	T	--	--	E	5	8				X			X
F	LUNLOC	T	CFFLG	A	--	--	M						X			X
G	LAT60[a]	H	DFLTT	F	--	--	P						X			X
H	PSUV		NFLT	L	--	--							X			X

a FBLO when NATF card current feedback plug is disconnected.

8-5.2 Software Testpoints

Testpoints TP1 and TP2 on the HFXB card in the SEM monitor the output of two digital-to-analog (d/a) converters. Each point presents a scaled representation of a software variable, selected via the thumbwheel switch (on the HFXA card, Figure 4-3). The digital value of the selected variable can be approximated by measuring the voltage at the testpoint and applying the indicated scaling. Refer to Table 8-3.

Table 8-3. Variables Monitored at Software Test Points TP1 & TP2 (HFXB)

TWS position	TP1 Variable	Scaling	TP2 Variable	Scaling
0	REF	a	ALPHA	c
1	REF	a	CEMF	a
2	REF	a	VAVE	a
3	REF	a	IFBK	a
4	REF	a	IAVE	a
5	CEMF	a	VAVE	a
6	IFBK	a	IAVE	a
7	IERR	a	BALERR	a
8	ALPHA	c	TMZERO	b
9	ALPHA	c	IPLL	a
A	FWDCMD	f	LATFL	f
B	IFBK	a	FWATTS	e
C	IFBK	a	RWATTS	e
D	FWATTS	e	RWATTS	e
E	TFJUNC	d	TRJUNC	d
F	TSINK	d	AFLO	g

Scaling Values

a ± 1.05 V / V
b - 10.5 V + 82.5 mV / bit
c - 10.5 V + 87.5 mV / degree of firing angle
d 70 mV / degree Celsius
e 2.58 V / kilowatt
f + 10.5 V = FF(H) = true ; - 10.5 V = 00(H) =false
g ± 82.5 mV / bit

8-5.3. Output Port Test Vectors

With the power converter off line, 21 control signals can be exercised to check internal and external logic. These signals are listed in Table 8-4.

To use this aid, HFXA switches SW1 and SW2 must be placed in their center and upper positions, respectively. Momentarily depress SW1 to step through the indicated test sequence. The test sequence number, in binary format, is displayed on the HFPB card.

Table 8-4. Output Port Test Vectors — I/O Test Mode

Test no.	Activated Output
00	None
01	Port 5 / bit 0: SFLT (LD1)
02	Port 5 / bit 1: PCA (LD2)
03	Port 5 / bit 2: STAT1 (LD3)
04	Port 5 / bit 3: STAT2 (LD4)
05	Port 5 / bit 4: FLTLI (RIL on door)
06	Port 5 / bit 5: TRIPA (dc breaker high speed trip)
07	Port 5 / bit 6: RLO (ac breaker shunt trip)
08	Port 5 / bit 7: DCSTA (dc breaker shunt trip)
09	Port C / bit 0: EXCTL (ext. ang. FF input)
10	Port C / bit 1: AVE2 (IFBK2 AVE. circuit)
11	Port C / bit 2: AVE3 (IFBK3 AVE. circuit)
12	Port C / bit 3: AVE4 (IFBK4 AVE. circuit)
13	Port C / bit 4: GOFWD (GO FWD. CMMD/ACK)
14	Port C / bit 5: FFRST (ext. ang. FF reset)
15	Port C / bit 6: CLRCT (clr. current bal. ct latch)
16	Port C / bit 7: REVI (rev. IFBK polarity)
17	Port D / bit 0: ALPUL (allow pulses)
18	Port D / bit 1: ALT (selects test osc.)
19	Port D / bit 2: CLRAC (clr. ac input latches)
20	Port D / bit 3: RST (resets IOC)
21	Port D / bit 4: RECAL (recall NOVRAM EEPROM data)
22	None

8-5.4 Circular List

NOTE

Software testpoints TP1 and TP2 are disabled when using the circular list.

The 77-mm SILCOMATIC Power Converter contains a circular list which has storage capacity for 1024 bytes, representing a maximum of four variables. The circular list can be run in three selectable software modules (firing logic, firing module, and dispatch) so that time-dependent variables can be monitored at the appropriate time. Sample rates are also selectable when running the list with the dispatch module.

To use the circular list, a terminal must be connected to the power converter and the operating system must be modified via the integral monitor program. Operating system commands and monitor commands are summarized in Tables 8-5 and 8-6, respectively.

Several parameters in computer memory must be altered to accomplish the following tasks:

1. Enable the circular list algorithm.
2. Specify the number and location of variables to be monitored.
3. Specify the list site (firing logic, firing module, or dispatch).
4. (Dispatch module only) Specify the sampling interval.

These parameters, their memory locations, and their range of values are tabulated in Table 8-7.

To use the circular list, the above-mentioned parameters must be set and then the hardware reset to enable the list. To print the list after triggering, dump the data beginning at CLARR and decrement 1024 times (printing each time) to read the entire list.

Table 8-5. 77-mm SILCOMATIC Operating System Commands (Entered with the Terminal)

Command	Explanation
S	Status. Dumps message queue.
V	Version. Prints system revision level
R	Reset. Causes system hardware reset
B	Bridge test. Initiates repeat of bridge test
M	Monitor. Switch to GE/DSD sideline monitor [a]
H	Help. Displays system commands

[a] WDB is the password for the monitor. If the monitor receives the incorrect password, the user cannot write to memory.

Table 8-6. GE/DSD Sideline Monitor Commands (Entered with the Terminal)

NOTE

Any characters other than legal hexadecimal numbers or the command characters shown below will be ignored by the computer

Command	Explanation
=	Sets the contents of the current address shown to the value entered after the "=" command.
+	Steps the terminal to the next address after the one currently shown.
-	Steps the terminal to the last address before the one currently shown.
CNTRL C	Return to 77-mm SILCOMATIC operating system
"RETURN"	Execute command

Table 8-7. Circular List Parameters

Address	Label	Possible Values	Explanation
3205	CLSEL		Enables circular list & defines site of circular list routine, which determines when data is sampled.
		00	Circular list disabled
		01	Firing logic module
		02	Dispatch module
		04	1-Hz sample rate
		08	10-Hz sample rate
		10	30-Hz sample rate
		20	60-Hz sample rate
		40	360-Hz sample rate
		80	Firing module
3207, 6	TGRPTR		Trigger data pointer
3208	-----	00	Non-inverted trigger(trigger on "1")
		01	Inverted trigger (trigger on "0")
3209	-----	-----	Trigger bit mask
320A	CLDLY	-----	Sample window (0.1s/bit)
320B	CLTAB		No. of variables to be monitored
		00	None
		01	One variable
		02	Two variables
		04	Four variables
320D, C	CBYT1	-----	Pointer to variable no. 1
320F, E	CBYT2	-----	Pointer to variable no. 2
3211, 10	CBYT3	-----	Pointer to variable no. 3
3213, 12	CBYT4	-----	Pointer to variable no. 4
3215, 14	CLARR	-----	Points to last byte written + 1

Notes

1. All addresses and values are in hexadecimal notation.
2. 16-bit (2-byte) addresses show MSB, then LSB.

CHAPTER 9
SPARE PARTS

9-1. RECOMMENDATIONS

The Drive Systems philosophy is to improve customer productivity and reduce equipment downtime by recommending to the customer that he:

1. maintain a stock of spare parts,
2. replace failures from spare stock,
3. replace spare stock with:
   a. warranty parts (during warranty period)
   b. use of equipment exchange plan (after warranty period)
   c. repair and return service (after warranty period)

There are two categories of spare parts: (1) start-up spares and (2) running spares. The start-up spares ensure that critical parts are available for equipment installation and tune-up. Running spares should be available during the normal life of the equipment to ensure minimum down-time.

9-2. DIRECTO-MATIC ELEMENT EXCHANGE SERVICE

This service is available for printed circuit boards and certain other assemblies. (Epoxy potted assemblies and oven cards are not included in this plan.) All printed wiring boards except PWB68A989554 are covered under this plan. Under this plan, a malfunctioning PWB or component can be returned and replaced by a like serviceable item from stock for only 40% of its original purchase price.

9-3. EMERGENCY SERVICE

In the event of an emergency when you have a vital operation shutdown for want of a part, General Electric will do its utmost to help you. Regardless of the time of day or night, company personnel will attempt to locate your part and expedite its delivery to you. To obtain this service, contact the General Electric Sales operation nearest your location.

9-4. PARTS DOCUMENTATION

After engineering has been completed, a Renewal Parts Quotation Booklet is customized for your equipment. This publication contains a list of all electrical devices and components used, a summary of these parts, device and major item internal parts list, and detailed parts bulletins for most devices. The Renewal Parts Quotation Booklet is also a valuable maintenance tool for the life of the equipment.

-- APPENDIX A --

HFXA	HFPB
A( )	( )A
B( )	( )B
C( )	( )C
D( )	( )C
E( )	( )E
F( )	( )F
G( )	( )G
H( )	( )H

Figure A-1. Identification of HFXA & HFPB LEDs

Table A-1. Computer Self-Test Sequence

Test no.	LEDs on HFXA	LEDs on HFPB	Test description
1	ALL	ALL	Lamp test
2	-	C	HFPB U4 PROM checksum test
3	-	D	HFPB U4 PROM starting address check
4	-	E	HFPB U5 PROM checksum test
5	-	F	HFPB U5 PROM starting address check
6	-	G	HFXA U10 PROM checksum test
7	-	H	HFXA U10 PROM starting address check
8	C	C	HFXA U11 PROM checksum test
9	C	D	HFXA U11 PROM starting address check
10	C	E	HFXB U4 PROM checksum test
11	C	F	HFXB U4 PROM starting address check
12	C	G	PROM compatibility check
	C	H	
13	D	C	HFPB U2 RAM test
	D	D	
14	D	E	HFPB U3 RAM test
	D	F	
15	D	G	HFXB U1 RAM test
	D	H	
16	E	C	HFXB U5 socket RAM test
	E	D	
	E	E	
	E	F	
	E	G	
	E	H	
	F	C	
	F	D	
17	F	E	DFXB NOVRAM test
	F	F	
18	F	G	Initialization

Table A-2. Terminal Messages -- Computer Self-Test

Test no.	Message
1 - 3	None (system inoperative)
4	E0-HFPB U5 PROM BAD
5	E1-HFPB U5 WRONG PROM
6	E2-HFXA U10 PROM BAD
7	E3-HFXA U10 WRONG PROM
8	E4-HFXA U11 PROM BAD
9	E5-HFXA U11 WRONG PROM
10	E6-HFXB U4 PROM BAD
11	E7-HFXB U4 WRONG PROM
12	E8-HFPB U5 PROM WRONG REV.
	E9-HFXA U10 PROM WRONG REV.
	EA-HFXA U11 PROM WRONG REV.
	EB-HFXB U4 PROM WRONG REV.
13	EC-HFPB U2 RAM FAILURE
14	ED-HFPB U3 RAM FAILURE
15	EE-HFXB U1 RAM FAILURE
16	EF-HFXB U5 RAM FAILURE
17	EG-DFXB NONVOLATILE RAM FAILURE
18	Q1-NONVOLATILE RAM NOT INITIALIZED INSERT BANANA PLUG THRU HFXB TP4 & TP5 THEN SEND CNTRL-S FROM TERMINAL
	ER-NO STO PULSE - CHECK JUMPER
	I3-NONVOLATILE RAM UPDATED
	EQ-DFXB CARD MEMORY FAULT
END	I0-SELFTEST COMPLETE: COMPUTER OK I1-SYSTEM INITIALIZED

Table A-3. Terminal Messages -- Bridge Test

```
A0-BRIDGE TEST FAULT:  NO PLL SYNC (CHECK AC POWER)
A0-BRIDGE TEST FAULT:  NO PLL INTERRUPT
A0-BRIDGE TEST FAULT:  MISSING PLL CODES
A0-BRIDGE TEST FAULT:  REVERSED PHASE ROTATION
A0-BRIDGE TEST FAULT:  SHORTED CELL - (P1,P2,P3,N1,N2,N3)
A0-BRIDGE TEST FAULT:
    FORWARD CELL NOT FIRING - (P1,P2,P3,N1,N2,N3)
A0-BRIDGE TEST FAULT:
    REVERSE CELL NOT FIRING - (P1,P2,P3,N1,N2,N3)
A0-BRIDGE TEST FAULT:  UNDEFINED BRIDGE FAULT
I2-BRIDGE TEST COMPLETE:  BRIDGE OK
```

Table A-4. Terminal Messages -- System Test

ES-SYSTEM TEST LOCKED OUT: AC PRESENT ON BRIDGE

Table A-5. Terminal Messages -- Operating Faults

```
B5 - INSTANTANEOUS OVER CURRENT DETECTED
B1 - MISSING PULSE DETECTED
B3 - CIRCULATING CURRENT FAULT DETECTED
B2 - DIAMETRIC FIRING FAULT DETECTED
B4 - COMMUTATING FAULT DETECTED
B7 - REVERSAL ERROR
B0 - BRIDGE CURRENT BALANCE FAULT
ET - FORWARD CELL NOT TAKING LOAD-(P1,P2,P3,N1,N2,N3) ON BRIDGE (1,2,3,4)
B0 - BRIDGE CURRENT BALANCE FAULT
EU - REVERSE CELL NOT TAKING LOAD-(P1,P2,P3,N1,N2,N3) ON BRIDGE (1,2,3,4)
```

Table A-6. Terminal Messages -- Power Supply Faults

EH-P5 SUPPLY BUS LOW
EI-P5 SUPPLY BUS HIGH
EJ-P15 SUPPLY BUS LOW
EK-P15 SUPPLY BUS HIGH
EN-N15 SUPPLY BUS LOW
EO-N15 SUPPLY BUS HIGH
EL-P28 SUPPLY BUS LOW
EM-P28 SUPPLY BUS HIGH
B6-POWER SUPPLY LVSL FAULT
EP-POWER SUPPLY BUSSES OK

Table A-7. Terminal Messages -- Logic Faults

C7-SILCO READY
C0-DC BREAKER OPEN
C1-IOC INTERLOCK SET
C2-THERMAL OVERLOAD FORWARD
C3-THERMAL OVERLOAD REVERSE
C4-PLL INTERLOCK SET
C5-HIGH AMBIENT TEMP. OR FEEDBACK OPEN
C6-POWER SUPPLY INTERLOCKSET

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